

DIFFERENTIAL EQUATIONS

From Local Change to Evolution

Calculus, Dynamics, and Waves

Extended mathematical problem

Complete ten-part paper

Extended paper — intended for several sittings

Calculators and electronic aids are not required.

Adapted from the *Differential Equations* notes based on lectures by M. G. Worster. Original notes by Dexter Chua.

Instructions and planned structure

This completed paper consists of one problem divided into the following ten parts.

- I. Differentiation, asymptotics, and local approximation;
- II. integration and multivariable differentiation;
- III. first-order linear differential equations;
- IV. nonlinear first-order equations and discrete dynamics;
- V. homogeneous second-order equations and phase space;
- VI. forced and equidimensional equations, recurrences, damping, and impulses;
- VII. series solutions near ordinary and singular points;
- VIII. gradients, Taylor forms, and stationary points;
- IX. differential systems and phase portraits;
- X. characteristics, waves, and diffusion.

Question numbers remain continuous across all ten parts. The result of any question may be used later, even if that question has not been answered, provided its number is cited clearly. Indications are optional. Definitions and notation are introduced when they first become necessary.

Prerequisites. The reader is assumed to know elementary algebra; real limits and their algebra; continuity; the usual elementary functions and their derivatives; the mean-value theorem; mathematical induction; the elementary integration of continuous functions; complex numbers, their real and imaginary parts and conjugates; convergence of real series and power series; monotone convergence for bounded monotone sequences; and elementary matrix notation, matrix-vector products, determinants, and linear independence and bases in $\mathbb{R}^n$. No prior course on differential equations, multivariable calculus, power-series methods, partial differential equations, or generalized functions is assumed: every later concept is defined in the paper, stated as a named admitted result, or proved in an earlier question. No result may be used unless it belongs to this declared background, has been stated as an admitted result, or has been proved in an earlier question.

Part I. — Differentiation, asymptotics, and local approximation

As in the original notes, the reader is assumed to know elementary single-variable real calculus: real limits and their algebra, continuity, the usual elementary functions and their derivatives, the mean-value theorem, and mathematical induction. The definitions and results emphasized by the course are nevertheless stated and developed here.

Admitted result. MVT Mean Value Theorem. Let $a, b \in \mathbb{R}$ satisfy $a < b$, and let $f: [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

A. — Derivatives and equations of evolution

Derivative. Let $I \subseteq \mathbb{R}$ be an open interval, let $f: I \rightarrow \mathbb{R}$, and let $x_0 \in I$. The *derivative* of f at x_0 is

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h},$$

provided this real limit exists. In that case, f is *differentiable at x_0* . It is differentiable on I if it is differentiable at every $x_0 \in I$. When f is differentiable on I , its *derivative function* is the function $f': I \rightarrow \mathbb{R}$ whose value at each $x \in I$ is the derivative of f at x . The right-hand and left-hand derivatives are defined by replacing $h \rightarrow 0$ with $h \rightarrow 0^+$ and $h \rightarrow 0^-$, respectively. Thus f is differentiable at x_0 precisely when both one-sided derivatives exist and are equal.

1)

- a) **DQPL** *Difference Quotient for a Power Law.* Let $p: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $p(x) = x^2$ for every $x \in \mathbb{R}$. Starting from the difference quotient, prove that $p'(x_0) = 2x_0$ for every $x_0 \in \mathbb{R}$.
- b) **ONDD** *One-sided Non-Differentiability Demonstration.* Let $a: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $a(x) = |x|$ for every $x \in \mathbb{R}$. Compute the right-hand and left-hand derivatives of a at 0 and prove that a is not differentiable there.
- c) **HDC** *Higher Derivative Conventions.* When f is differentiable, its derivative function is also written

$$\frac{df}{dx} = f'(x) = \frac{d}{dx} f(x).$$

If f' is differentiable, set $f'' = (f')'$ and write $d^2f/dx^2 = f''(x)$. Let $f: I \rightarrow \mathbb{R}$ be differentiable, put $J = \{t \in \mathbb{R} : 2t \in I\}$, and let $g: J \rightarrow \mathbb{R}$ be defined by $g(t) = f(2t)$. Use the difference quotient to prove $g'(t) = 2f'(2t)$ for every $t \in J$.

Differential equations. Let $I \subseteq \mathbb{R}$ be an interval. A *differential equation* is an equation imposed on an unknown function $x: I \rightarrow \mathbb{R}$ and some of its derivatives. A *solution on I* is a function possessing the required derivatives and satisfying the equation at every $t \in I$. When the argument is time t , write

$$\dot{x}(t) = \frac{dx}{dt}(t), \quad \ddot{x}(t) = \frac{d^2x}{dt^2}(t).$$

Initial conditions. An *initial condition* prescribes one or more values of the unknown function and its derivatives at a specified time.

Linear equations. For equations involving an unknown twice differentiable function $x: \mathbb{R} \rightarrow \mathbb{R}$, call an equation *linear* if it can be written

$$a_2(t)\ddot{x}(t) + a_1(t)\dot{x}(t) + a_0(t)x(t) = b(t) \quad (t \in \mathbb{R}),$$

where $a_0, a_1, a_2, b: \mathbb{R} \rightarrow \mathbb{R}$ are prescribed functions independent of x .

Nonlinear equations. An equation that cannot be written in the linear form above is *nonlinear*. For example, an equation containing a non-zero multiple of $x(t)^2$, or a product $x(t)\dot{x}(t)$, cannot be written in that form and is therefore nonlinear.

2) Let $m \in \mathbb{R}$ satisfy $m > 0$. In a one-dimensional Newtonian model, $t \in \mathbb{R}$ denotes time, a twice differentiable function $x: \mathbb{R} \rightarrow \mathbb{R}$ gives the particle's position, and its force is the function $F_x: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$F_x(t) = m\ddot{x}(t) \quad (t \in \mathbb{R}).$$

a) **PADE** *Prescribed Acceleration Differential Equation.* Suppose that $F_x(t) = t^2 - 1$ for every $t \in \mathbb{R}$. Prove that the twice differentiable functions $x: \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$m\ddot{x}(t) = t^2 - 1 \quad (t \in \mathbb{R})$$

are exactly the functions

$$x(t) = \frac{t^4}{12m} - \frac{t^2}{2m} + c_1t + c_0 \quad (t \in \mathbb{R}),$$

where $c_0, c_1 \in \mathbb{R}$.

b) **PDNDE** *Position-Dependent Nonlinear Differential Equation.* Suppose instead that $F_x(t) = x(t)^2 - 1$ for every $t \in \mathbb{R}$, so that

$$m\ddot{x}(t) = x(t)^2 - 1 \quad (t \in \mathbb{R}).$$

Determine all constant solutions $x: \mathbb{R} \rightarrow \mathbb{R}$.

c) **ICSU** *Initial Conditions Select Uniquely.* Let $x_0, v_0 \in \mathbb{R}$. Among the solutions in part (a), find the unique one satisfying $x(0) = x_0$ and $\dot{x}(0) = v_0$.

Indication. — For the converse statement in part (a), subtract the displayed candidate from an arbitrary solution and use the standard consequence of the mean-value theorem that a function with zero second derivative is affine.

Write $\mathbb{N} = \{1, 2, 3, \dots\}$ and $\mathbb{N}_0 = \{0, 1, 2, \dots\}$.

Real sequences. A *real sequence* is a map $F: \mathbb{N}_0 \rightarrow \mathbb{R}$, whose value at $n \in \mathbb{N}_0$ is denoted by F_n .

Difference equations. A *difference equation* is a relation between terms of an unknown sequence. A *solution* is a sequence satisfying that relation for every index for which it is stated.

3) Consider the Fibonacci difference equation

$$F_{n+2} - F_{n+1} - F_n = 0 \quad (n \in \mathbb{N}_0).$$

a) **FRCS** *Fibonacci Recurrence Computes Successors.* For the solution with $F_0 = 0$ and $F_1 = 1$, compute $F_2, F_3, \dots, F_7$.

b) **TRIVFS** *Two Recurrence Initial Values Fix the Sequence.* Let $F, G: \mathbb{N}_0 \rightarrow \mathbb{R}$ be two solutions. Prove by induction that $F_0 = G_0$ and $F_1 = G_1$ imply $F_n = G_n$ for every $n \in \mathbb{N}_0$.

c) **RANC** *Recurrence Alone is Not Complete.* Exhibit two distinct solutions of the difference equation that have the same value $F_0 = 0$.

The explicit formula for the Fibonacci solution will be derived in Part VI. These first examples also illustrate a recurring method: one may guess a candidate solution, but the candidate becomes a solution only after its domain, differentiability, and governing equation have been checked.

B. — Small and bounded asymptotic orders

Deleted neighbourhoods. Let $I \subseteq \mathbb{R}$ be an interval, let $x_0 \in \mathbb{R} \cup \{-\infty, +\infty\}$ be such that I contains points arbitrarily close to x_0 in the relevant direction, and let $f, g: I \rightarrow \mathbb{R}$. All limits below are taken through points $x \in I$. A *deleted neighbourhood* of a finite x_0 consists of the points satisfying $0 < |x - x_0| < \delta$ for some $\delta > 0$; at $+\infty$ it consists of the points satisfying $x > M$ for some $M \in \mathbb{R}$, and at $-\infty$ it is defined analogously.

Neighbourhoods. A *neighbourhood* of a finite $x_0 \in \mathbb{R}$ is a set containing an open interval that contains x_0 . A function is *defined near* x_0 if it is defined on some neighbourhood of x_0 ; more generally, a property of a function is said to hold *near* x_0 if it holds throughout some neighbourhood of x_0 .

Small order. Suppose that $g(x) \neq 0$ throughout some deleted neighbourhood of x_0 . We write

$$f(x) = o(g(x)) \quad \text{as } x \rightarrow x_0$$

if

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = 0.$$

Thus $o(g)$ denotes the class of all functions f satisfying this limit, and the conventional equality $f = o(g)$ is the membership assertion

$$f \in \{h : \lim_{x \rightarrow x_0} h(x)/g(x) = 0\}.$$

Informally, $o(g)$ is negligible compared with g .

Bounded order. With the same conventions, we write

$$f(x) = O(g(x)) \quad \text{as } x \rightarrow x_0$$

if there exist $C \in \mathbb{R}$ with $C \geq 0$ and a deleted neighbourhood of x_0 throughout which

$$|f(x)| \leq C|g(x)|.$$

This definition remains meaningful without the non-vanishing assumption on g , and $f = O(g)$ means that f belongs to the class of functions bounded in magnitude by a constant multiple of g .

4)

- a) **SROZ** *Square-Root Order near Zero.* On the interval $(0, \infty)$, prove that $x = o(\sqrt{x})$ as $x \rightarrow 0^+$.
- b) **SRGI** *Square Root Growth at Infinity.* On the interval $(0, \infty)$, prove that $\sqrt{x} = o(x)$ as $x \rightarrow +\infty$.
- c) **LSOB** *Local Sine Order Bound.* Using the elementary inequality $|\sin u| \leq |u|$ for every $u \in \mathbb{R}$, prove that $\sin(2x) = O(x)$ as $x \rightarrow 0$.
- d) **SBI** *Sine Bounded at Infinity.* Prove that $\sin(2x) = O(1)$ as $x \rightarrow +\infty$, although $\sin(2x)$ has no limit as $x \rightarrow +\infty$.
- e) **VRGBR** *Vanishing Ratio Gives a Bounded Ratio.* Let $f, g: I \rightarrow \mathbb{R}$ satisfy the hypotheses in the definition of small order. Prove that

$$f(x) = o(g(x)) \implies f(x) = O(g(x)) \quad (x \rightarrow x_0).$$

5) In every part, all functions are real-valued and all asymptotic statements refer to the same limit $x \rightarrow x_0$.

- a) **SRAC** *Scaling Rules for Asymptotic Classes.* Let $\lambda \in \mathbb{R}$. Prove that $f = o(g)$ implies $\lambda f = o(g)$, and that $f = O(g)$ implies $\lambda f = O(g)$.
- b) **SAR** *Sum of Asymptotic Remainders.* Prove that $f_1 = o(g)$ and $f_2 = o(g)$ imply $f_1 + f_2 = o(g)$. Prove the analogous assertion with O in place of o .

- c) **PROB** *Product Remainder Order Bound.* Prove that $f = O(g)$ and $p = O(q)$ imply $fp = O(gq)$. Deduce that, as $h \rightarrow 0$, the relations $r(h) = o(h)$ and $s(h) = O(h)$ imply $r(h)s(h) = o(h^2)$.
- d) **CRCL** *Composition Remainder Control Lemma.* Let $\delta > 0$, let $k: (-\delta, \delta) \rightarrow \mathbb{R}$ satisfy $k(h) \rightarrow 0$ and $k(h) = O(h)$ as $h \rightarrow 0$, and let ε be a real-valued function defined on an open interval containing 0, with $\varepsilon(u) \rightarrow 0$ as $u \rightarrow 0$. Prove

$$k(h)\varepsilon(k(h)) = o(h) \quad (h \rightarrow 0).$$

- 6) Let $I \subseteq \mathbb{R}$ be an open interval, let $x_0 \in I$, and let $f: I \rightarrow \mathbb{R}$.
- a) **DILA** *Derivative Implies Linear Approximation.* Suppose that f is differentiable at x_0 . Starting from the derivative limit, prove

$$f(x_0 + h) = f(x_0) + f'(x_0)h + o(h) \quad (h \rightarrow 0).$$

- b) **LADD** *Linear Approximation Determines the Derivative.* Conversely, suppose that $A \in \mathbb{R}$ and a function r , defined near 0, satisfy

$$f(x_0 + h) = f(x_0) + Ah + r(h), \quad r(h) = o(h) \quad (h \rightarrow 0).$$

Prove that f is differentiable at x_0 and $f'(x_0) = A$.

- c) **LAC** *Linear Approximation Coefficient.* Prove that at most one number $A \in \mathbb{R}$ can occur as the coefficient of h in an expansion of the form in part (b).
- d) **QEO** *Quadratic Error Order.* For $f(x) = x^2$ on $\mathbb{R}$, show directly that

$$f(x_0 + h) = f(x_0) + 2x_0h + h^2.$$

Prove that $h^2 = O(h^2)$ and $h^2 = o(h)$ as $h \rightarrow 0$.

C. — Product, composition, and repeated differentiation

- 7) Let $I \subseteq \mathbb{R}$ be an open interval, let $x_0 \in I$, and let $u, v: I \rightarrow \mathbb{R}$ be differentiable at x_0 .

- a) **PRE** *Product Remainder Expansion.* Introduce functions r_u, r_v , defined near 0, such that $r_u(h) = o(h)$, $r_v(h) = o(h)$, and

$$u(x_0 + h) = u(x_0) + hu'(x_0) + r_u(h),$$

$$v(x_0 + h) = v(x_0) + hv'(x_0) + r_v(h).$$

Multiply these two identities.

- b) **CTRC** *Cross-Term Remainder Control.* Using Question 5, prove that all terms in the product from part (a), other than the constant term and the terms linear in h , have sum $o(h)$.
- c) **DPR** *Derivative Product Rule.* Apply Question 6 to prove that

$$(uv)'(x_0) = u'(x_0)v(x_0) + u(x_0)v'(x_0).$$

- 8) Let $I, J \subseteq \mathbb{R}$ be open intervals, let $g: I \rightarrow J$, let $F: J \rightarrow \mathbb{R}$, and let $x_0 \in I$. Suppose that g is differentiable at x_0 and F is differentiable at $y_0 = g(x_0)$.

- a) **ICI** *Inner Composition Increment.* Let k be defined on a sufficiently small neighbourhood of 0 by $k(h) = g(x_0 + h) - g(x_0)$. Prove that

$$k(h) = hg'(x_0) + o(h), \quad k(h) = O(h), \quad k(h) \rightarrow 0 \quad (h \rightarrow 0).$$

- b) **ORF** *Outer Remainder Function.* For every sufficiently small $s \in \mathbb{R}$ with $y_0 + s \in J$, let

$$\varepsilon(s) = \begin{cases} \frac{F(y_0 + s) - F(y_0) - F'(y_0)s}{s}, & s \neq 0, \\ 0, & s = 0. \end{cases}$$

Prove that $\varepsilon(s) \rightarrow 0$ as $s \rightarrow 0$.

- c) **CRB** *Composed Remainder Bound.* Prove

$$F(g(x_0 + h)) - F(g(x_0)) = F'(y_0)k(h) + k(h)\varepsilon(k(h)).$$

Use Question 5(d) to obtain a linear approximation in h .

- d) **DCR** *Derivative Chain Rule.* Conclude that $F \circ g$ is differentiable at x_0 and

$$(F \circ g)'(x_0) = F'(g(x_0))g'(x_0).$$

Repeated derivatives. Let $I \subseteq \mathbb{R}$ be an open interval and let $f: I \rightarrow \mathbb{R}$. For $n \in \mathbb{N}_0$, set $f^{(0)} = f$ and, whenever the required derivatives exist, define $f^{(n+1)} = (f^{(n)})'$.

Factorials. Define $0! = 1$ and $n! = 1 \cdot 2 \cdots n$ for $n \in \mathbb{N}$.

Binomial coefficients. If $n \in \mathbb{N}_0$ and $r \in \mathbb{N}_0$ satisfies $0 \leq r \leq n$, define

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

9) Let $I \subseteq \mathbb{R}$ be an open interval and let $u, v: I \rightarrow \mathbb{R}$ possess every derivative that occurs below.

- a) **ZOPC** *Zeroth and Onefold Product Cases.* Verify that

$$(uv)^{(n)} = \sum_{r=0}^n \binom{n}{r} u^{(r)} v^{(n-r)} \quad (L_n)$$

holds for $n = 0$ and $n = 1$.

- b) **DIH** *Differentiated Induction Hypothesis.* Let $n \in \mathbb{N}$. Assume (L_n) and differentiate its two sides. Use the product rule to obtain two finite sums representing $(uv)^{(n+1)}$.
- c) **PCR** *Pascal Combination after Reindexing.* Reindex one of the sums from part (b), and prove directly from the factorial definition that

$$\binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r} \quad (1 \leq r \leq n).$$

Combine the two sums, treating the terms with $r = 0$ and $r = n + 1$ separately.

- d) **GLDFP** *General Leibniz Derivative Formula for Products.* Use induction to prove that, for every $n \in \mathbb{N}_0$ and all n -times differentiable functions $u, v: I \rightarrow \mathbb{R}$,

$$(uv)^{(n)}(x) = \sum_{r=0}^n \binom{n}{r} u^{(r)}(x) v^{(n-r)}(x) \quad (x \in I).$$

Indication.— In part (c), replace $r+1$ by a new index in the sum containing $u^{(r+1)}v^{(n-r)}$ before combining coefficients.

D. — Taylor polynomials and formal series

Taylor polynomial. Let $n \in \mathbb{N}_0$, let $I \subseteq \mathbb{R}$ be open, let $x_0 \in I$, and let $f: I \rightarrow \mathbb{R}$ possess derivatives through order n at x_0 . The *Taylor polynomial of order n about x_0* is

$$T_{n,x_0}f(x) = \sum_{r=0}^n \frac{f^{(r)}(x_0)}{r!} (x - x_0)^r \quad (x \in \mathbb{R}).$$

Admitted result. TTPR Taylor Theorem with Peano Remainder. Let $n \in \mathbb{N}_0$, let $I \subseteq \mathbb{R}$ be an open interval, let $x_0 \in I$, and let $f: I \rightarrow \mathbb{R}$. Suppose that the derivatives $f^{(0)}, f^{(1)}, \dots, f^{(n)}$ exist on a neighbourhood of x_0 and that $f^{(n)}$ is continuous at x_0 . Then, as $h \rightarrow 0$ with $x_0 + h \in I$,

$$f(x_0 + h) = \sum_{r=0}^n \frac{f^{(r)}(x_0)}{r!} h^r + o(h^n).$$

If, in addition, $f^{(n+1)}$ exists and is bounded on a neighbourhood of x_0 , the remainder after the displayed finite sum is $O(h^{n+1})$.

The statement is local: it describes f only on a neighbourhood of x_0 , and for one fixed finite n it asserts nothing about the convergence of the infinite Taylor series or about the equality of its sum with f .

10) Retain the notation and hypotheses of the admitted Taylor theorem.

- a) **FTPC Finite Taylor Polynomial Construction.** Prove that, for every integer $k \in \mathbb{N}_0$ with $0 \leq k \leq n$,

$$(T_{n,x_0}f)^{(k)}(x_0) = f^{(k)}(x_0).$$

- b) **EVTF Equivalent Variable Taylor Form.** Put $x = x_0 + h$ and rewrite the admitted result as an expansion of $f(x)$ as $x \rightarrow x_0$. Write the remainder in both the o form and, under the additional hypothesis, the O form.
- c) **CTE Cubic Taylor Expansion.** Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^3$. Compute $T_{3,1}f(x)$ and verify directly that it equals $f(x)$ for every $x \in \mathbb{R}$.

Taylor series. Let $I \subseteq \mathbb{R}$ be open, let $x_0 \in I$, and let $f: I \rightarrow \mathbb{R}$ possess derivatives of every order at x_0 . The *Taylor series of f about x_0* is the formal power series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n.$$

The Taylor series about $x_0 = 0$ is also called the *Maclaurin series*. The word *formal* means that this definition alone asserts only that the coefficients are defined; it asserts neither convergence of the numerical series at a specified x nor equality of its sum with $f(x)$.

11)

- a) **PTSF Polynomial Taylor Series is Finite.** Let $m \in \mathbb{N}_0$, let $a_0, \dots, a_m \in \mathbb{R}$, and let the polynomial function $p: \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$p(x) = \sum_{r=0}^m a_r x^r \quad (x \in \mathbb{R}).$$

Prove that the Taylor series of p about 0 has no non-zero terms after degree m and is equal to $p(x)$ for every $x \in \mathbb{R}$.

- b) **MTRF Maclaurin Terms for a Rational Function.** Let $f: \mathbb{R} \setminus \{1\} \rightarrow \mathbb{R}$ be defined by $f(x) = (1 - x)^{-1}$. Prove that $f^{(n)}(0) = n!$ for every $n \in \mathbb{N}_0$ and hence determine its formal Maclaurin series.

E. — Quotients with vanishing numerator and denominator

An expression $f(x)/g(x)$ has the *indeterminate form* $0/0$ at $x_0 \in \mathbb{R}$ when $f(x) \rightarrow 0$ and $g(x) \rightarrow 0$ as $x \rightarrow x_0$. The phrase does not assert that the quotient has a limit.

12) Let $I \subseteq \mathbb{R}$ be an open interval, let $x_0 \in I$, and let $f, g: I \rightarrow \mathbb{R}$ be differentiable at x_0 . Suppose $f(x_0) = g(x_0) = 0$ and $g'(x_0) \neq 0$.

- a) **FLQE** *First-order Linear Quotient Expansions.* Use Question 6 to write expansions of $f(x_0 + h)$ and $g(x_0 + h)$ to first order, with remainders in $o(h)$.
- b) **PNNZD** *Punctured-Neighbourhood Non-Zero Denominator.* Prove that there exists $\delta > 0$ such that $g(x_0 + h) \neq 0$ whenever $0 < |h| < \delta$.
- c) **LQDL** *Local Quotient Derivative Limit.* Divide the expansions from part (a) by h , justify the quotient using part (b), and prove

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \frac{f'(x_0)}{g'(x_0)}.$$

Indication.— For part (b), use $g(x_0 + h)/h \rightarrow g'(x_0) \neq 0$ and the fact that a function with a non-zero limit is non-zero sufficiently near the limit point.

Admitted result. **GDQLR** *General Derivative Quotient Limit Rule.* Let $x_0 \in \mathbb{R}$, let $\delta > 0$, put $I = (x_0 - \delta, x_0 + \delta)$, and let $f, g: I \rightarrow \mathbb{R}$ be continuous. Suppose that f and g are differentiable at every $x \in I \setminus \{x_0\}$, that

$$f(x_0) = g(x_0) = 0, \quad g'(x) \neq 0 \quad (x \in I \setminus \{x_0\}),$$

and that there exists $L \in \mathbb{R}$ such that

$$\lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)} = L.$$

Then $g(x) \neq 0$ throughout some deleted neighbourhood of x_0 and

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = L.$$

13)

- a) **SQR** *Sine Quotient Recovery.* Apply Question 12 to the functions $f, g: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \sin x$ and $g(x) = x$ to recover

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

- b) **CSOQL** *Cosine Second-Order Quotient Limit.* For $g(x) = x^2$, show that $g'(0) = 0$, so that the hypothesis $g'(x_0) \neq 0$ of Question 12 fails at $x_0 = 0$. Use the admitted general rule and part (a) to prove

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}.$$

- c) **RDQE** *Repeated Derivative Quotient Evaluation.* Use the admitted general rule and part (b) to prove

$$\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3} = \frac{1}{6}.$$

- d) **HRVA** *Hypothesis Review before Valid Application.* Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $g(x) = x$ for every $x \in \mathbb{R}$, $f(0) = 0$, and

$$f(x) = x^2 \sin(1/x) \quad (x \in \mathbb{R} \setminus \{0\}).$$

Prove directly that $f(x)/g(x) \rightarrow 0$ as $x \rightarrow 0$. Compute $f'(x)/g'(x)$ for $x \neq 0$ and prove that it has no limit as $x \rightarrow 0$, although the original quotient converges.

End of Part I

Part II. — Integration and multivariable differentiation

This part constructs integration from finite sums and then extends the local viewpoint of Part I to functions of several real variables. The elementary algebra of real limits and continuous functions remains available.

A. — Integrals and their evaluation

Uniform partitions. Let $a, b \in \mathbb{R}$ satisfy $a < b$, and let $f: [a, b] \rightarrow \mathbb{R}$. For each positive integer $N \in \mathbb{N}$, put

$$\Delta x_N = \frac{b-a}{N}, \quad x_{n,N} = a + n\Delta x_N \quad (0 \leq n \leq N).$$

Left Riemann sums. With the notation above, define the *left Riemann sum*

$$S_N(f; a, b) = \sum_{n=0}^{N-1} f(x_{n,N}) \Delta x_N.$$

The term indexed by n is the signed area of a rectangle of width Δx_N and height $f(x_{n,N})$: it is positive where f is positive and negative where f is negative.

Definite integrals. For a continuous function f , the finite real limit

$$\lim_{N \rightarrow \infty} S_N(f; a, b)$$

exists by the result stated immediately below. Call this limit the *definite integral* of f over $[a, b]$ and write it as $\int_a^b f(x) dx$. Define $\int_b^a f(x) dx = -\int_a^b f(x) dx$ and $\int_a^a f(x) dx = 0$. The complete theory of Riemann integration is, as in the original notes, part of Analysis I rather than this course; in particular, this paper does not use left sums as a definition for arbitrary discontinuous functions.

*Admitted result. **BRIP** Basic Riemann Integral Properties.* Let $a, b, c \in \mathbb{R}$ satisfy $a \leq c \leq b$. For every continuous $f: [a, b] \rightarrow \mathbb{R}$, the definite integral $\int_a^b f(x) dx$ exists. If $f, g: [a, b] \rightarrow \mathbb{R}$ are continuous and $\alpha, \beta \in \mathbb{R}$, then

$$\int_a^b (\alpha f(x) + \beta g(x)) dx = \alpha \int_a^b f(x) dx + \beta \int_a^b g(x) dx,$$

and $\int_a^b f = \int_a^c f + \int_c^b f$. If $m, M \in \mathbb{R}$ satisfy $m \leq f(x) \leq M$ for every $x \in [a, b]$, then $m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$.

14) Let $a, b \in \mathbb{R}$ satisfy $a < b$.

- a) **URSP** *Uniform Riemann Sum Partition.* Let $f: [a, b] \rightarrow \mathbb{R}$. Prove that $x_{0,N} = a$, $x_{N,N} = b$, and $x_{n+1,N} - x_{n,N} = \Delta x_N$ for every $N \in \mathbb{N}$ and every integer n with $0 \leq n < N$.
- b) **LSRF** *Linear Sum Recovers Formula.* Let $f: [a, b] \rightarrow \mathbb{R}$ be defined by $f(x) = x$. Using $\sum_{n=0}^{N-1} n = N(N-1)/2$, compute $S_N(f; a, b)$ and prove that

$$\int_a^b x dx = \frac{b^2 - a^2}{2}.$$

- c) **RSEO** *Rectangle Sum Error Order.* Suppose that $f: [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b) , and that there is a real number $M \geq 0$ such that $|f'(x)| \leq M$ for every $x \in (a, b)$. Use the mean-value theorem to prove that, for every $N \in \mathbb{N}$ and every integer n with $0 \leq n < N$,

$$\left| \int_{x_{n,N}}^{x_{n+1,N}} (f(x) - f(x_{n,N})) dx \right| \leq \frac{M}{2} (\Delta x_N)^2.$$

Deduce that

$$\left| \int_a^b f(x) \, dx - S_N(f; a, b) \right| \leq \frac{M(b-a)}{2} \Delta x_N,$$

which makes precise the local error $O((\Delta x_N)^2)$ and the total error $O(\Delta x_N)$.

Accumulation functions. Let $a, b \in \mathbb{R}$ satisfy $a < b$, let $f: [a, b] \rightarrow \mathbb{R}$ be continuous, and define $F: [a, b] \rightarrow \mathbb{R}$ by

$$F(x) = \int_a^x f(t) \, dt \quad (x \in [a, b]).$$

At an endpoint, a derivative below means the appropriate one-sided derivative.

15) Let $x \in [a, b]$ and let $h \in \mathbb{R} \setminus \{0\}$ be such that $x + h \in [a, b]$.

- a) **IALI** *Integral Accumulation as a Local Increment.* Use additivity and orientation of the integral to prove

$$\frac{F(x+h) - F(x)}{h} = \frac{1}{h} \int_x^{x+h} f(t) \, dt.$$

- b) **CAML** *Continuity gives an Average Mean Limit.* Let $\varepsilon \in \mathbb{R}$ satisfy $\varepsilon > 0$. Use continuity of f at x and the integral bound in BRIP to prove that, whenever h is sufficiently close to 0,

$$\left| \frac{1}{h} \int_x^{x+h} f(t) \, dt - f(x) \right| \leq \varepsilon.$$

Treat both signs of h .

- c) **FTCD** *Fundamental Theorem Connecting Differentiation.* Deduce that F is continuous on $[a, b]$, differentiable on (a, b) , and satisfies $F'(x) = f(x)$ for every $x \in (a, b)$, with the corresponding one-sided identities at a and b . This is the Fundamental Theorem of Calculus in the form used in the original notes.

Interior points. Let $I \subseteq \mathbb{R}$ be an interval. Call $x \in I$ an *interior point* of I if some open interval containing x is contained in I ; thus the interior points are exactly the points of I that are not endpoints.

Antiderivatives. Let $I \subseteq \mathbb{R}$ be an interval and let $f: I \rightarrow \mathbb{R}$ be continuous. An *antiderivative* of f on I is a differentiable function $A: I \rightarrow \mathbb{R}$ such that $A'(x) = f(x)$ for every interior point $x \in I$, with one-sided derivatives at any included endpoint. The notation $\int f(x) \, dx = A(x) + C$ denotes the family of all antiderivatives, where $C \in \mathbb{R}$ is a constant of integration.

16) Let $a, b \in I$ and let $g: J \rightarrow I$ be differentiable on an interval $J \subseteq \mathbb{R}$.

- a) **ORL** *Orientation Reversal for Limits.* Prove that

$$\frac{d}{dx} \int_x^b f(t) \, dt = -f(x)$$

for every interior point $x \in I$ by first reversing the limits.

- b) **VLDF** *Variable-Limit Derivative Formula.* Use Question 15 and the chain rule from Question 8 to prove

$$\frac{d}{dx} \int_a^{g(x)} f(t) \, dt = f(g(x))g'(x)$$

at every interior point $x \in J$.

- c) **ACFC** *Antiderivatives Carry a Free Constant.* Let $A_1, A_2: I \rightarrow \mathbb{R}$ be antiderivatives of f . Use the mean-value theorem to prove that $A_1 - A_2$ is constant on I and hence justify the constant C in the indefinite-integral notation.

In the substitution formula below, one conventionally writes $u = g(x)$ and $du = g'(x)dx$, so that the integral takes the form $\int f(u) du$.

17)

- a) **ICS** *Integral Chain Substitution.* Let $I, J \subseteq \mathbb{R}$ be intervals, let $g: I \rightarrow J$ be differentiable, and let $f: J \rightarrow \mathbb{R}$ be continuous. If $A: J \rightarrow \mathbb{R}$ is an antiderivative of f , use the chain rule to prove

$$\int f(g(x))g'(x) dx = A(g(x)) + C.$$

- b) **QSSE** *Quadratic Square root Substitution Example.* On any open interval on which $x - x^2 > 0$, evaluate

$$\int \frac{1 - 2x}{\sqrt{x - x^2}} dx$$

by using $u = x - x^2$. Differentiate your answer to check it.

- c) **STSI** *Shifted Trigonometric Substitution Integral.* On the interval $(0, 2)$, put $x - 1 = \sin \theta$ with $\theta \in (-\pi/2, \pi/2)$. Derive

$$\int \sqrt{2x - x^2} dx = \frac{1}{2} \arcsin(x - 1) + \frac{1}{2}(x - 1)\sqrt{2x - x^2} + C.$$

- d) **HSCP** *Hyperbolic Substitution Choice Practice.* For each expression $\sqrt{1 - x^2}$, $1 + x^2$, $\sqrt{x^2 - 1}$, $\sqrt{1 + x^2}$, and $1 - x^2$, exhibit a substitution chosen from

$$x = \sin \theta, \quad x = \tan \theta, \quad x = \cosh u, \quad x = \sinh u, \quad x = \tanh u,$$

together with a real interval on which it is one-to-one, and prove the resulting identity that removes the square root or simplifies the expression.

18) Let $I \subseteq \mathbb{R}$ be an interval and let $u, v: I \rightarrow \mathbb{R}$ be continuously differentiable.

- a) **IBPF** *Integration By-Parts Formula.* Integrate the product-rule identity $(uv)' = u'v + uv'$ and prove

$$\int_a^b u(x)v'(x) dx = [u(x)v(x)]_a^b - \int_a^b v(x)u'(x) dx$$

for every $a, b \in I$. Deduce the corresponding indefinite formula.

- b) **EII** *Exponential Improper Integral.* For a continuous function $q: [0, \infty) \rightarrow \mathbb{R}$, define $\int_0^\infty q(x) dx$ to be $\lim_{R \rightarrow \infty} \int_0^R q(x) dx$ when this finite limit exists. Use integration by parts and the limit $Re^{-R} \rightarrow 0$ to prove

$$\int_0^\infty xe^{-x} dx = 1.$$

- c) **LAI** *Logarithm Antiderivative Identity.* Use integration by parts to prove, for every $x \in (0, \infty)$,

$$\int \log x dx = x \log x - x + C.$$

B. — Partial and total differentiation

Open sets. Let $n \in \mathbb{N}$. A set $U \subseteq \mathbb{R}^n$ is *open* if, for every $x = (x_1, \dots, x_n) \in U$, there is $\varepsilon > 0$ such that every $y = (y_1, \dots, y_n) \in \mathbb{R}^n$ satisfying

$$(y_1 - x_1)^2 + \dots + (y_n - x_n)^2 < \varepsilon^2$$

belongs to U . Thus an open set contains, with each of its points, all points sufficiently close to it. In particular, every open interval is an open subset of $\mathbb{R}$.

Neighbourhoods in several variables. These local notions extend to points of $\mathbb{R}^n$: a *neighbourhood* of $p \in \mathbb{R}^n$ is a set containing an open set that contains p , and a function is *defined near* p if it is defined on some neighbourhood of p .

Partial derivatives. Let $U \subseteq \mathbb{R}^2$ be open, let $f: U \rightarrow \mathbb{R}$, and let $(x_0, y_0) \in U$. The *partial derivative with respect to the first variable* is

$$f_x(x_0, y_0) = \frac{\partial f}{\partial x}(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h},$$

provided the real limit exists. Define $f_y(x_0, y_0)$ analogously by holding x_0 fixed. Iterated derivatives use the order indicated by the subscripts: $f_{xy} = (f_x)_y$ and $f_{yx} = (f_y)_x$.

19) Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by $f(x, y) = x^2 + y^3 + e^{xy^2}$ for every $(x, y) \in \mathbb{R}^2$.

- a) **PDQ** *Partial Derivative Quotients.* Starting with the defining limit, compute $f_x(x, y)$ and $f_y(x, y)$ for every $(x, y) \in \mathbb{R}^2$.
- b) **EMPC** *Exponential Multivariable Partial Computation.* Compute $f_{xx}(x, y)$ and $f_{xy}(x, y)$ for every $(x, y) \in \mathbb{R}^2$.
- c) **MPDN** *Mixed Partial Derivative Notation.* Compute $f_{yx}(x, y)$ and verify directly that it equals $f_{xy}(x, y)$.

Admitted result. **SCECP** *Schwarz Commutation for Existing Continuous Partial.* Let $U \subseteq \mathbb{R}^2$ be open and let $f: U \rightarrow \mathbb{R}$. If the second partial derivatives f_{xy} and f_{yx} exist and are continuous on U , then $f_{xy}(x, y) = f_{yx}(x, y)$ for every $(x, y) \in U$.

Differentiability in two variables. Let $U \subseteq \mathbb{R}^2$ be open, let $f: U \rightarrow \mathbb{R}$, and let $(x_0, y_0) \in U$. Say that f is *differentiable at* (x_0, y_0) if there exist $A, B \in \mathbb{R}$ and a function r defined near $(0, 0)$ such that, for all sufficiently small $(h, k) \in \mathbb{R}^2$,

$$f(x_0 + h, y_0 + k) - f(x_0, y_0) = Ah + Bk + r(h, k)$$

and $r(h, k)/(|h| + |k|) \rightarrow 0$ as $(h, k) \rightarrow (0, 0)$ with $(h, k) \neq (0, 0)$. A function is *differentiable on* U if it is differentiable at every point of U . Question 20(a) identifies A and B with the two partial derivatives.

Total differential. With the notation above, the linear expression

$$df = A dx + B dy$$

is called the *total differential* at (x_0, y_0) .

20) Let $f: U \rightarrow \mathbb{R}$ be differentiable at $(x_0, y_0) \in U$.

- a) **RPLI** *Rectangular Path Linear Increment.* Show from the definition that the coefficients satisfy $A = f_x(x_0, y_0)$ and $B = f_y(x_0, y_0)$. Hence prove

$$\delta f = f_x(x_0, y_0)\delta x + f_y(x_0, y_0)\delta y + o(|\delta x| + |\delta y|).$$

- b) **TDAE** *Total Differential Approximation Error.* Suppose additionally that f_x and f_y exist near (x_0, y_0) and are continuous there. Split the increment along the rectangular path $(x_0, y_0) \rightarrow (x_0 + \delta x, y_0) \rightarrow (x_0 + \delta x, y_0 + \delta y)$ and use one-variable linear approximations to recover the formula in part (a).

- c) **MCR** *Multivariable Chain Rule.* Let $J \subseteq \mathbb{R}$ be an open interval, let $x, y: J \rightarrow \mathbb{R}$ be differentiable at $t_0 \in J$, and suppose $(x(t), y(t)) \in U$ for every $t \in J$ and $(x(t_0), y(t_0)) = (x_0, y_0)$. Substitute their one-variable linear approximations into that of f to prove

$$\left. \frac{d}{dt} f(x(t), y(t)) \right|_{t=t_0} = f_x(x(t_0), y(t_0))x'(t_0) + f_y(x(t_0), y(t_0))y'(t_0).$$

- d) **CPR** *Chain rule along a Plane Route.* Now suppose that f is differentiable on U . Let $y: J \rightarrow \mathbb{R}$ be differentiable and suppose $(x, y(x)) \in U$ for every $x \in J$. Deduce

$$\frac{d}{dx} f(x, y(x)) = f_x(x, y(x)) + f_y(x, y(x))y'(x)$$

at every $x \in J$.

21) Let $V = (0, \infty) \times \mathbb{R}$, and let $T: V \rightarrow \mathbb{R}^2$ be defined by $T(r, \theta) = (r \cos \theta, r \sin \theta)$, and let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be differentiable. Put $F = f \circ T$.

- a) **PCC** *Polar Coordinate Chain rule.* Apply Question 20 with one variable fixed at a time to prove

$$F_\theta(r, \theta) = -r \sin \theta f_x(T(r, \theta)) + r \cos \theta f_y(T(r, \theta)).$$

- b) **RAD** *Radial and Angular Derivatives.* Derive a corresponding expression for $F_r(r, \theta)$. For $(x, y) = T(r, \theta)$, solve the resulting two linear equations for $f_x(x, y)$ and $f_y(x, y)$ in terms of $F_r(r, \theta)$ and $F_\theta(r, \theta)$.

Differentiability in three variables. Let $U \subseteq \mathbb{R}^3$ be open, let $F: U \rightarrow \mathbb{R}$, and let $(x_0, y_0, z_0) \in U$. Say that F is *differentiable at* (x_0, y_0, z_0) if there exist $A, B, C \in \mathbb{R}$ and a function r defined near $(0, 0, 0)$ such that, for all sufficiently small $(h, k, \ell) \in \mathbb{R}^3$,

$$F(x_0 + h, y_0 + k, z_0 + \ell) - F(x_0, y_0, z_0) = Ah + Bk + C\ell + r(h, k, \ell),$$

where

$$\frac{r(h, k, \ell)}{|h| + |k| + |\ell|} \rightarrow 0 \quad \text{as } (h, k, \ell) \rightarrow (0, 0, 0),$$

through non-zero triples. It is differentiable on U if it is differentiable at every point of U . As in two variables, setting two increments equal to zero shows that $A = F_x$, $B = F_y$, and $C = F_z$ at the point in question.

Implicit differentiation. Let $U \subseteq \mathbb{R}^3$ be open, let $F: U \rightarrow \mathbb{R}$ be differentiable, let $c \in \mathbb{R}$, and let $D \subseteq \mathbb{R}^2$ be open. Suppose that a differentiable function $z: D \rightarrow \mathbb{R}$ satisfies $(x, y, z(x, y)) \in U$ and

$$F(x, y, z(x, y)) = c \quad ((x, y) \in D).$$

Thus this equation defines the graph of z as part of the level surface of F with value c . The formulas derived below divide by F_z , so they apply only at points where $F_z \neq 0$.

22)

- a) **IDCE** *Implicit Differentiation Computation Example.* Let $F: \mathbb{R}^3 \rightarrow \mathbb{R}$ be given by $F(x, y, z) = xy^2 + yz^2 + xz^5$. At a point on the level surface $F = 5$ where z is locally a differentiable function of (x, y) , differentiate with respect to x while holding y fixed and prove that, whenever $2yz + 5xz^4 \neq 0$,

$$z_x = -\frac{y^2 + z^5}{2yz + 5xz^4}.$$

- b) **IFD** *Implicit Formula Derivation.* In the general setting above, fix y and use the three-variable linear approximation of F , together with the one-variable linear approximation of $x \mapsto z(x, y)$, to prove that wherever $F_z(x, y, z(x, y)) \neq 0$,

$$z_x = -\frac{F_x}{F_z}.$$

Derive similarly

$$z_y = -\frac{F_y}{F_z},$$

with every partial derivative evaluated at $(x, y, z(x, y))$.

- c) **NZCI** *Non-Zero Condition Interpretation.* Give an example of a differentiable $F: \mathbb{R}^2 \rightarrow \mathbb{R}$ and a level set $F(x, z) = c$ that cannot be represented as a differentiable graph $z = z(x)$ near a point where $F_z = 0$.

C. — Differentiation under an integral sign

Admitted result. **CII** *Continuous Integral Interchange.* Let $a, b, c_0, c_1 \in \mathbb{R}$ satisfy $a < b$ and $c_0 < c_1$. If $f: [a, b] \times [c_0, c_1] \rightarrow \mathbb{R}$ and its partial derivative f_c are continuous, then the function $I: [c_0, c_1] \rightarrow \mathbb{R}$ defined by $I(c) = \int_a^b f(y, c) dy$ is differentiable on (c_0, c_1) and

$$I'(c) = \int_a^b f_c(y, c) dy \quad (c \in (c_0, c_1)),$$

with the corresponding one-sided derivatives at the endpoints.

23) Let $J \subseteq \mathbb{R}$ be an open interval, let $a, b, c: J \rightarrow \mathbb{R}$ be differentiable, and suppose $a(x) < b(x)$ for every $x \in J$. Let $W \subseteq \mathbb{R}^2$ be an open set containing every $(y, c(x))$ with $x \in J$ and $a(x) \leq y \leq b(x)$, and let $f: W \rightarrow \mathbb{R}$ have continuous partial derivatives.

- a) **FPDI** *Fixed-limit Parameter Differentiation under Integration.* For fixed real numbers $\alpha < \beta$, let $K(c) = \int_\alpha^\beta f(y, c) dy$ for each c for which the segment $[\alpha, \beta] \times \{c\}$ lies in W . Compute $K'(c)$ using CII.
- b) **CULR** *Changing Upper Limit Rule.* Fix $x_0 \in J$. By continuity, choose $d \in \mathbb{R}$ and a neighbourhood of x_0 on which $a(x) < d < b(x)$. Let H be defined on a neighbourhood of $(b(x_0), c(x_0))$ by $H(u, v) = \int_d^u f(y, v) dy$. Compute H_u and H_v , and apply the multivariable chain rule to $H(b(x), c(x)) - H(a(x), c(x))$.
- c) **GLE** *General Leibniz Expression.* Deduce the Leibniz integral rule

$$\begin{aligned} \frac{d}{dx} \int_{a(x)}^{b(x)} f(y, c(x)) dy &= f(b(x), c(x))b'(x) - f(a(x), c(x))a'(x) \\ &\quad + c'(x) \int_{a(x)}^{b(x)} f_c(y, c(x)) dy. \end{aligned}$$

- d) **PGIE** *Parameterized Gaussian Integral Examples.* For every $\lambda \in (0, \infty)$, let

$$I(\lambda) = \int_0^1 e^{-\lambda y^2} dy, \quad J(\lambda) = \int_0^\lambda e^{-\lambda y^2} dy.$$

Use the Leibniz rule to prove

$$I'(\lambda) = -\int_0^1 y^2 e^{-\lambda y^2} dy, \quad J'(\lambda) = e^{-\lambda^3} - \int_0^\lambda y^2 e^{-\lambda y^2} dy.$$

End of Part II

Part III. — First-order linear differential equations

This part explains the special role of exponentials, introduces the precise classification of first-order equations, and develops homogeneous, inhomogeneous, and variable-coefficient solution methods.

A. — Exponentials as eigenfunctions

Exponential functions with base. For each real number $a > 0$, the *exponential function with base a* is the function $E_a: \mathbb{R} \rightarrow (0, \infty)$ defined by $E_a(x) = a^x$.

Natural exponential. The *natural exponential* is the unique differentiable function $\exp: \mathbb{R} \rightarrow (0, \infty)$ satisfying

$$\exp'(x) = \exp(x) \quad (x \in \mathbb{R}), \quad \exp(0) = 1.$$

Write $e = \exp(1)$ and $e^x = \exp(x)$. The natural exponential is strictly increasing and maps $\mathbb{R}$ onto $(0, \infty)$.

Natural logarithm. The inverse of the natural exponential is the *natural logarithm* $\log: (0, \infty) \rightarrow \mathbb{R}$, also written $\ln$. These existence and inverse-function facts belong to the elementary calculus background of the original notes.

24) Let $a \in (0, \infty)$.

- a) **BPDP** *Base Power Derivative Proportionality.* Starting from the derivative definition and the identity $a^{x+h} = a^x a^h$, prove that

$$E'_a(x) = \lambda_a E_a(x), \quad \lambda_a = \lim_{h \rightarrow 0} \frac{a^h - 1}{h},$$

for every $x \in \mathbb{R}$. Deduce that λ_a depends on a but not on x .

- b) **NECL** *Natural Exponential Characterizing Limit.* Apply part (a) with $a = e$ and use $\exp'(0) = 1$ to prove $\lambda_e = 1$.
- c) **GENE** *General Exponential through Natural Exponentiation.* From $a = e^{\log a}$, derive

$$a^x = e^{x \log a} \quad (x \in \mathbb{R}),$$

and use the chain rule to prove $\lambda_a = \log a$.

Complex-valued differentiation. Let $I \subseteq \mathbb{R}$ be an open interval. A function $f: I \rightarrow \mathbb{C}$ is differentiable at $x \in I$ when the difference quotient has a limit in $\mathbb{C}$; equivalently, its real and imaginary parts are differentiable there.

Complex exponential. For $\lambda = u + iv \in \mathbb{C}$, where $u, v \in \mathbb{R}$, and $x \in \mathbb{R}$, define

$$e^{\lambda x} = e^{ux} (\cos(vx) + i \sin(vx)).$$

Eigenfunctions of the derivative operator. A non-zero differentiable function $f: I \rightarrow \mathbb{C}$ is an *eigenfunction of the differential operator* $\frac{d}{dx}$ with *eigenvalue* $\lambda \in \mathbb{C}$ when $f'(x) = \lambda f(x)$ for every $x \in I$. Here non-zero means not identically zero.

25) Let $I \subseteq \mathbb{R}$ be a non-empty open interval and let $\lambda \in \mathbb{C}$.

- a) **CEDE** *Complex Exponential Derivative Eigenvalue.* Differentiate the displayed real-and-imaginary-part definition to prove

$$\frac{d}{dx} e^{\lambda x} = \lambda e^{\lambda x} \quad (x \in \mathbb{R}).$$

- b) **OECG** *Only Exponential Candidates are General.* Let $f: I \rightarrow \mathbb{C}$ be differentiable and satisfy $f' = \lambda f$. Let $g: I \rightarrow \mathbb{C}$ be defined by $g(x) = f(x)e^{-\lambda x}$, prove $g'(x) = 0$, and deduce that

$$f(x) = C e^{\lambda x} \quad (x \in I)$$

for some $C \in \mathbb{C}$. Include the zero solution by allowing $C = 0$.

- c) **ICVS** *Initial Condition Value Selection.* Given $x_0 \in I$ and $y_0 \in \mathbb{C}$, determine the unique solution of $f' = \lambda f$ on I satisfying $f(x_0) = y_0$.

B. — Homogeneous equations and two limiting constructions

First-order linear equations. Let $I \subseteq \mathbb{R}$ be an interval and let $a, b, c: I \rightarrow \mathbb{R}$. An equation for an unknown differentiable function $y: I \rightarrow \mathbb{R}$ is *first-order linear* if it has the form

$$a(x)y'(x) + b(x)y(x) = c(x) \quad (x \in I),$$

that is, if y and y' occur only to the first power and are not multiplied together.

Homogeneous equations. The equation above is *homogeneous* if $c(x) = 0$ for every $x \in I$, equivalently if $y = 0$ is a solution.

Forced equations. The equation is *inhomogeneous*, or *forced*, when it is not homogeneous; c is then the *forcing term*.

Constant coefficients. The equation has *constant coefficients* if a and b are constant.

Initial conditions for first-order equations. An *initial condition* for such an equation specifies a value $y(x_0) = y_0$ for some $x_0 \in I$ and $y_0 \in \mathbb{R}$.

Boundary conditions. The same algebraic condition imposed at an endpoint is also called a boundary condition in the original notes.

- 26)** Assume in this question that $p: I \rightarrow \mathbb{R}$ is nonconstant and that $f: I \rightarrow \mathbb{R}$ is not identically zero.

- a) **LHC** *Linear Homogeneous Classification.* For each equation

$$y' + p(x)y = 0, \quad y' + p(x)y = f(x), \quad 5y' - 3y = 0, \quad yy' = 1,$$

prove or disprove that it is linear, prove or disprove that it is homogeneous, and determine whether its coefficients are constant. The functions p and f are the prescribed nonconstant and non-zero functions above.

- b) **CAER** *Characteristic Ansatz Equation Reduction.* For the equation $5y' - 3y = 0$ on $\mathbb{R}$, substitute the ansatz $y(x) = e^{mx}$ with $m \in \mathbb{R}$ and derive the algebraic equation $5m - 3 = 0$.
- c) **GSHS** *General Solution of a Homogeneous Scalar equation.* Rewrite the equation as $y' = (3/5)y$ and apply Question 25 to prove that every real-valued solution is

$$y(x) = Ae^{3x/5} \quad (x \in \mathbb{R})$$

for a unique $A \in \mathbb{R}$.

- d) **ICUS** *Initial Condition Uniquely Selects.* Let $x_0, y_0 \in \mathbb{R}$. Find the value of A for which the solution in part (c) satisfies $y(x_0) = y_0$.

Admitted result. **ELC** *Euler Limit Characterization.* For every $z \in \mathbb{R}$,

$$\lim_{n \rightarrow \infty} \left(1 + \frac{z}{n}\right)^n = e^z,$$

where the expression is considered for all sufficiently large positive integers n so that $1 + z/n > 0$.

- 27)** Let $y_{\text{in}} \in \mathbb{R}$ and let $h \in (0, \infty)$. Put $x_n = nh$ for every $n \in \mathbb{N}_0$, and let $(y_n)_{n \in \mathbb{N}_0}$ be a real sequence with $y_0 = y_{\text{in}}$. Approximate the equation $5y' - 3y = 0$ at x_n by replacing $y'(x_n)$ with $(y_{n+1} - y_n)/h$.

- a) **FED** *Forward Euler Discretization.* Derive the recurrence

$$y_{n+1} = \left(1 + \frac{3h}{5}\right)y_n \quad (n \in \mathbb{N}_0).$$

- b) **RPF** *Recurrence Product Formula.* Prove by induction that the unique sequence with initial value y_{in} is

$$y_n = \left(1 + \frac{3h}{5}\right)^n y_{\text{in}} \quad (n \in \mathbb{N}_0).$$

- c) **CEL** *Continuous Exponential Limit.* Fix $x \in (0, \infty)$. For every $N \in \mathbb{N}$, put $h_N = x/N$ and let $(y_j^{(N)})_{j \in \mathbb{N}_0}$ be the sequence from parts (a) and (b) with step h_N and initial value $y_0^{(N)} = y_{\text{in}}$. Prove that its value at the endpoint $x_N = Nh_N = x$ is

$$y_N^{(N)} = y_{\text{in}} \left(1 + \frac{3x}{5N}\right)^N,$$

and use ELC to show that $y_N^{(N)} \rightarrow y_{\text{in}} e^{3x/5}$ as $N \rightarrow \infty$.

Formal power series. For the next question, a *formal power series* is an expression $\sum_{n=0}^{\infty} a_n x^n$ manipulated by its coefficients; convergence is not asserted. Two formal power series are equal precisely when their coefficients of every power x^n , $n \in \mathbb{N}_0$, are equal.

Formal derivative. The *formal derivative* of a formal power series is defined by

$$\left(\sum_{n=0}^{\infty} a_n x^n\right)' = \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n.$$

Admitted result. **EPSI** *Exponential Power Series Identity.* For every $z \in \mathbb{R}$, the real series

$$\sum_{n=0}^{\infty} \frac{z^n}{n!}$$

converges and has sum e^z .

28) Seek a formal solution $y = \sum_{n=0}^{\infty} a_n x^n$ of $5y' - 3y = 0$, where $(a_n)_{n \in \mathbb{N}_0}$ is a sequence in $\mathbb{R}$.

- a) **VPSC** *Vanishing Power Series Coefficients.* Substitute the series and show that equality to the zero series is equivalent to

$$5(n+1)a_{n+1} - 3a_n = 0 \quad (n \in \mathbb{N}_0).$$

- b) **CRC** *Coefficient Recurrence Computation.* Prove by induction that

$$a_n = \frac{1}{n!} \left(\frac{3}{5}\right)^n a_0 \quad (n \in \mathbb{N}_0),$$

using $0! = 1$ and $(n+1)! = (n+1)n!$.

- c) **EPSR** *Exponential Power Series Recovery.* Using EPSI, recover the actual solution $y(x) = a_0 e^{3x/5}$ for every $x \in \mathbb{R}$.

C. — Forcing, superposition, and integrating factors

Linear differential operators. Let $I \subseteq \mathbb{R}$ be an interval and let L be a linear differential operator on differentiable functions $I \rightarrow \mathbb{R}$, meaning that

$$L[\alpha u + \beta v] = \alpha L[u] + \beta L[v]$$

for all differentiable $u, v: I \rightarrow \mathbb{R}$ and all $\alpha, \beta \in \mathbb{R}$. An operator *annihilates* a function when it maps that function to the zero function.

Particular solutions. For $L[y] = f$, a *particular solution* or *particular integral* is one specific solution y_p .

Complementary functions. For $L[y] = f$, the *complementary function* y_c is the general solution of $L[y_c] = 0$.

29) Let $f: I \rightarrow \mathbb{R}$ and suppose $y_p: I \rightarrow \mathbb{R}$ satisfies $L[y_p] = f$.

- a) **LODS** *Linear Operator Distribution over Sums.* If $y_c: I \rightarrow \mathbb{R}$ satisfies $L[y_c] = 0$, use linearity to prove $L[y_p + y_c] = f$.
- b) **DHS** *Difference is a Homogeneous Solution.* If $y: I \rightarrow \mathbb{R}$ is any solution of $L[y] = f$, prove that $L[y - y_p] = 0$.
- c) **GSFS** *General Solution Follows Superposition.* Deduce that the complete solution set of $L[y] = f$ is

$$\{y_p + y_c : L[y_c] = 0\}.$$

30) Consider the equation $5y' - 3y = 10$ on $\mathbb{R}$.

- a) **CEP** *Constant Equilibrium Particular solution.* Seek a constant particular solution and prove that $y_p = -10/3$ is the unique constant solution.
- b) **FCGS** *Forced Constant General Solution.* Use Questions 26 and 29 to prove that every solution is

$$y(x) = -\frac{10}{3} + Ae^{3x/5} \quad (x \in \mathbb{R})$$

for a unique $A \in \mathbb{R}$.

- c) **BCAF** *Boundary Condition Applied to the Full solution.* Let $x_0, y_0 \in \mathbb{R}$. Determine A from $y(x_0) = y_0$. Compute also the value A_c that would result from imposing the same condition on the complementary term alone, and prove that $A - A_c = \frac{10}{3}e^{-3x_0/5}$ for all x_0, y_0 .

31) Let time satisfy $t \in [0, \infty)$. Let $a, b: [0, \infty) \rightarrow \mathbb{R}$ be differentiable abundance functions, let $a_0 \in [0, \infty)$, and let $k_a, k_b \in (0, \infty)$ satisfy $k_a \neq k_b$. Suppose

$$a' = -k_a a, \quad b' = k_a a - k_b b, \quad a(0) = a_0, \quad b(0) = 0.$$

- a) **PADS** *Parent Abundance Decay Solution.* Solve the first initial-value problem and obtain $a(t) = a_0 e^{-k_a t}$ for every $t \geq 0$.
- b) **DFES** *Daughter Forced Exponential Solution.* Substitute part (a) into the equation for b , seek a particular solution $Ce^{-k_a t}$, and determine C .
- c) **ICFR** *Initial Condition Fixes the Response.* Add the complementary solution $De^{-k_b t}$, impose $b(0) = 0$, and prove

$$b(t) = \frac{k_a a_0}{k_b - k_a} (e^{-k_a t} - e^{-k_b t}) \quad (t \geq 0).$$

- d) **IRFC** *Isotope Ratio Formula for Chronology.* Assuming $a_0 > 0$, calculate $R(t) = b(t)/a(t)$. Given a measured ratio R , solve the resulting equation for the age t in terms of R , k_a , and k_b .

Standard form. Let $I \subseteq \mathbb{R}$ be an open interval and let $a, b, c: I \rightarrow \mathbb{R}$ be continuous, with $a(x) \neq 0$ for every $x \in I$. Dividing $a(x)y' + b(x)y = c(x)$ by $a(x)$ gives the *standard form*

$$y' + p(x)y = f(x), \quad p = \frac{b}{a}, \quad f = \frac{c}{a}.$$

Integrating factor. Fix $x_0 \in I$. The function $\mu: I \rightarrow (0, \infty)$ defined by

$$\mu(x) = \exp \left(\int_{x_0}^x p(s) \, ds \right) \quad (x \in I)$$

is called an *integrating factor*.

32)

- a) **IFDE** *Integrating Factor Differential Equation.* Use Question 15 and the chain rule to prove $\mu'(x) = p(x)\mu(x)$ for every $x \in I$.
- b) **PEDF** *Product Exact Derivative Form.* Multiply the standard equation by μ and use the product rule to prove

$$(\mu y)' = \mu f.$$

- c) **GIFF** *General Integrating Factor Formula.* Integrate between x_0 and $x \in I$ and prove that every solution is

$$y(x) = \mu(x)^{-1} \left(C + \int_{x_0}^x \mu(s)f(s) \, ds \right)$$

for a unique $C \in \mathbb{R}$. Prove the converse by differentiation.

33) Consider

$$xy'(x) + (1-x)y(x) = 1$$

on an interval not containing 0.

- a) **SFDD** *Standard Form Domain Decision.* Divide by x and determine $p(x) = 1/x - 1$ and $f(x) = 1/x$. Show that any interval on which the standard form is defined must avoid 0, so that $(0, \infty)$ and $(-\infty, 0)$ are the maximal such intervals.
- b) **IFC** *Integrating Factor Computation.* On $(0, \infty)$, compute an integrating factor $\mu(x) = xe^{-x}$.
- c) **GSBE** *General Solution Before the Endpoint.* Apply Question 32 to prove that every solution on $(0, \infty)$ has the form

$$y(x) = \frac{Ce^x - 1}{x}$$

for some $C \in \mathbb{R}$.

- d) **FBC** *Finite Boundary Condition.* Determine the unique $C \in \mathbb{R}$ for which $y(x)$ has a finite limit as $x \rightarrow 0^+$, and use Question 12 to compute that limit.

End of Part III

Part IV. — Nonlinear first-order equations and discrete dynamics

This part develops two solvable nonlinear classes and then studies evolution geometrically. Every explicit formula is accompanied by its domain conditions, since division during separation can otherwise discard or add solutions.

A. — Separable and exact equations

Differential forms. Let $I \subseteq \mathbb{R}$ be an interval, let $D \subseteq \mathbb{R}^2$ be open, and let $P, Q: D \rightarrow \mathbb{R}$ be continuous. For a differentiable function $y: I \rightarrow \mathbb{R}$ whose graph lies in D , the notation

$$P(x, y) dx + Q(x, y) dy = 0$$

means $P(x, y(x)) + Q(x, y(x))y'(x) = 0$ for every $x \in I$.

Separable equations. Let $I, J \subseteq \mathbb{R}$ be intervals. The equation above is *separable* on a region if it can be written $q(y) dy = p(x) dx$ for functions of one variable $q: J \rightarrow \mathbb{R}$ and $p: I \rightarrow \mathbb{R}$.

34) Let $I, J \subseteq \mathbb{R}$ be open intervals, let $p: I \rightarrow \mathbb{R}$ and $q: J \rightarrow \mathbb{R}$ be continuous, and let $P_0: I \rightarrow \mathbb{R}$ and $Q_0: J \rightarrow \mathbb{R}$ be antiderivatives of p and q , respectively.

- a) **SDF** *Separable Differential Form.* Prove that a differentiable function $y: I \rightarrow J$ satisfies $q(y(x))y'(x) = p(x)$ for every $x \in I$ if and only if

$$\frac{d}{dx}(Q_0(y(x)) - P_0(x)) = 0 \quad (x \in I).$$

- b) **IISC** *Implicit Integral Solution Criterion.* Deduce that every solution satisfies

$$Q_0(y(x)) = P_0(x) + C \quad (x \in I)$$

for some $C \in \mathbb{R}$, and prove conversely that every differentiable $y: I \rightarrow J$ satisfying this identity solves the equation.

- c) **NSE** *Nonlinear Separation Example.* Let $I \subseteq \mathbb{R}$ be an open interval on which $x^2 \neq 3$, and suppose $y: I \rightarrow \mathbb{R}$ is differentiable and satisfies

$$(x^2 y - 3y)y' - 2xy^2 = 4x.$$

Rearrange without dividing by y to obtain

$$\frac{yy'}{2 + y^2} = \frac{2x}{x^2 - 3},$$

and derive

$$\frac{1}{2} \log(2 + y(x)^2) = \log|x^2 - 3| + C.$$

- d) **DRC** *Domain Restriction Check.* Exponentiate the preceding relation to obtain $\sqrt{2 + y(x)^2} = A|x^2 - 3|$ for some $A > 0$. Differentiate this relation to check the original equation. Show also that the division performed in part (c) is valid throughout I and would fail at any point where $x^2 = 3$.

Domains. A *domain* is a non-empty open subset $D \subseteq \mathbb{R}^2$ in which any two points can be joined by a continuous curve lying in D .

Exact differential forms. Let $D \subseteq \mathbb{R}^2$ be a domain and let $P, Q: D \rightarrow \mathbb{R}$. The differential form $P dx + Q dy$ is *exact* on D if there is a differentiable function $\Phi: D \rightarrow \mathbb{R}$ satisfying $\Phi_x = P$ and $\Phi_y = Q$ throughout D ; Φ is called a *potential*, and the solution curves are then the level sets $\Phi(x, y) = C$ for $C \in \mathbb{R}$.

Simple connectivity. A *curve* in D is a continuous map $\gamma: [0, 1] \rightarrow D$; it is *closed* if $\gamma(0) = \gamma(1)$. A domain $D \subseteq \mathbb{R}^2$ is *simply connected* if every closed curve in D can be continuously contracted to a point through curves lying in D . A disc is simply connected, whereas an annulus is not. In three dimensions a sphere is simply connected, whereas a torus is not.

35) Assume $P, Q: D \rightarrow \mathbb{R}$ have continuous first partial derivatives.

a) **EDC** *Exact Differential Components.* Suppose $P dx + Q dy = d\Phi$ for a function $\Phi: D \rightarrow \mathbb{R}$ having continuous second partial derivatives. Prove that $P = \Phi_x$ and $Q = \Phi_y$ throughout D .

b) **MEC** *Mixed Equality Condition.* Apply SCECP to prove the necessary condition

$$P_y(x, y) = Q_x(x, y) \quad ((x, y) \in D).$$

c) **HDO** *Holed Domain Obstruction.* On $D = \mathbb{R}^2 \setminus \{(0, 0)\}$, let

$$P(x, y) = -\frac{y}{x^2 + y^2}, \quad Q(x, y) = \frac{x}{x^2 + y^2}.$$

Verify $P_y = Q_x$. Parameterize the unit circle by $(x(t), y(t)) = (\cos t, \sin t)$ for $t \in [0, 2\pi]$ and show that a potential would have to increase by 2π around this closed curve. Deduce that equality of mixed derivatives is not sufficient on every domain.

Admitted result. **SCES** *Simply-Connected Exactness Sufficiency.* Let $D \subseteq \mathbb{R}^2$ be an open simply-connected domain, and let $P, Q: D \rightarrow \mathbb{R}$ have continuous first partial derivatives. If $P_y = Q_x$ throughout D , then there exists a single-valued differentiable function $\Phi: D \rightarrow \mathbb{R}$ satisfying $\Phi_x = P$ and $\Phi_y = Q$.

36) On $\mathbb{R}^2$, put $P(x, y) = 2x - 3y^2$ and $Q(x, y) = 6y^2 - 6xy$.

a) **EVC** *Exactness Verification Calculation.* Compute P_y and Q_x , and use SCES to establish exactness.

b) **PRF** *Potential Reconstruction Function.* Integrate $\Phi_x = P$ with respect to x and write $\Phi(x, y) = x^2 - 3xy^2 + h(y)$ for some differentiable function $h: \mathbb{R} \rightarrow \mathbb{R}$. Use $\Phi_y = Q$ to determine h up to an additive constant.

c) **ILS** *Implicit Level Solutions.* Prove that the solution curves of

$$6y(y - x)y' + 2x - 3y^2 = 0$$

lie on the level sets

$$x^2 - 3xy^2 + 2y^3 = C,$$

where $C \in \mathbb{R}$, and verify the statement by differentiating along a differentiable branch $y = y(x)$.

B. — Trajectories, isoclines, and stability

Solution curves. Let $I \subseteq \mathbb{R}$ be an interval, let $D \subseteq \mathbb{R}^2$, and let $f: D \rightarrow \mathbb{R}$. A differentiable function $y: I \rightarrow \mathbb{R}$ whose graph lies in D and which solves $y' = f(t, y)$ determines a *solution curve*, or *trajectory*, $t \mapsto (t, y(t))$.

Isoclines. For $d \in \mathbb{R}$, an *isocline of slope d* is the set of points $(t, y) \in D$ satisfying $f(t, y) = d$. Several isoclines, marked with their slopes, indicate the directions of the trajectories without solving the equation.

37) Let $I \subseteq \mathbb{R}$ be an interval and let $y: I \rightarrow \mathbb{R}$ be a differentiable solution of $y' = t(1 - y^2)$.

a) **EFSC** *Explicit Family of Solution Curves.* On any subinterval where $|y| \neq 1$, separate variables and use

$$\frac{1}{1 - y^2} = \frac{1}{2} \left(\frac{1}{1 + y} + \frac{1}{1 - y} \right)$$

to derive

$$\frac{1}{2} \log \left| \frac{1 + y}{1 - y} \right| = \frac{t^2}{2} + C.$$

Show how this gives $y(t) = (A - e^{-t^2})/(A + e^{-t^2})$ for some $A \in \mathbb{R} \setminus \{0\}$, wherever the denominator is non-zero.

- b) **MSS** *Missing Stationary Solutions.* Determine the constant solutions lost when dividing by $1 - y^2$ and check them directly in the differential equation.
- c) **QSP** *Qualitative Slope Partition.* For $t > 0$, determine the sign of y' in each of the regions $y < -1$, $-1 < y < 1$, and $y > 1$. Show also that every solution has a horizontal tangent at $t = 0$ when $0 \in I$.
- d) **ICD** *Isocline Curve Derivation.* For $d \in \mathbb{R}$, derive the isocline equation $y^2 = 1 - d/t$ when $t \neq 0$, and determine the pairs (t, d) for which it has real solutions. Treat $t = 0$ separately.

Equilibria. For $y' = f(t, y)$, an *equilibrium* or *fixed point* is a real number a such that $f(t, a) = 0$ for every allowed t ; then $y(t) = a$ is a constant solution. Assume here that the solutions under discussion are defined for every $t \geq 0$.

Stability. An equilibrium a is *stable* if, for every $\epsilon > 0$, there exists $\delta > 0$ such that $|y(0) - a| < \delta$ implies $|y(t) - a| < \epsilon$ for every $t \geq 0$.

Attraction. An equilibrium a is *attracting* if there exists $\delta > 0$ such that $|y(0) - a| < \delta$ implies $y(t) \rightarrow a$ as $t \rightarrow \infty$.

Asymptotic stability. An equilibrium is *asymptotically stable* if it is both stable and attracting. It is *unstable* if it is not stable.

Linearisation. A *linearisation* keeps only terms linear in a small perturbation.

38) Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ have a continuous second partial derivative with respect to y , let $a \in \mathbb{R}$, and suppose $f(t, a) = 0$ for every $t \geq 0$.

- a) **EPE** *Equilibrium Perturbation Expansion.* Let $y: [0, \infty) \rightarrow \mathbb{R}$ be a differentiable solution and let $\varepsilon: [0, \infty) \rightarrow \mathbb{R}$ be defined by $y(t) = a + \varepsilon(t)$. For each fixed $t \geq 0$, first use Taylor expansion in the scalar increment $\eta \rightarrow 0$ to prove

$$f(t, a + \eta) = f_y(t, a)\eta + O(\eta^2).$$

Substitute $\eta = \varepsilon(t)$ whenever the perturbation is small and deduce

$$\varepsilon'(t) = f_y(t, a)\varepsilon(t) + O(\varepsilon(t)^2)$$

with the remainder understood in this local sense.

- b) **LEM** *Linearized Evolution Model.* Discard the quadratic remainder in part (a) and write the resulting linearised equation $\varepsilon' = f_y(t, a)\varepsilon$.
- c) **CSULE** *Contrasting Stability of Upper and Lower Equilibria.* Apply this procedure to $f(t, y) = t(1 - y^2)$. At $a = 1$ and $a = -1$ solve the two linearised equations and obtain respectively $\varepsilon(t) = \varepsilon(0)e^{-t^2}$ and $\varepsilon(t) = \varepsilon(0)e^{t^2}$. Use the explicit family and slope information in Question 37 to confirm that 1 is asymptotically stable and -1 is unstable.

Indication.— A growing linearised perturbation is only evidence of local instability: the approximation is valid while $|\varepsilon|$ is small, and the exact sign argument of part (c) is what proves the conclusion.

Autonomous scalar equations. An equation $y' = f(y)$ is *autonomous* because its right-hand side has no explicit dependence on time.

Admitted result. **SAUS** *Scalar Autonomous Uniqueness for Solutions.* Let $J \subseteq \mathbb{R}$ be an open interval and let $f: J \rightarrow \mathbb{R}$ be continuously differentiable. If two differentiable functions on the same time interval solve $y' = f(y)$ and have the same value at one time, then they agree throughout that time interval.

39) Let $J \subseteq \mathbb{R}$ be an open interval, let $f: J \rightarrow \mathbb{R}$ be twice continuously differentiable, and let $a \in J$ satisfy $f(a) = 0$.

- a) **ASDC** *Autonomous Stability Derivative Criterion.* Show that the linearised perturbation equation is $\varepsilon' = f'(a)\varepsilon$ and solve it. Use continuity of f' to determine the sign of $(y - a)f(y)$ near a , and use SAUS to show that a nonconstant solution cannot cross a . Deduce asymptotic stability when $f'(a) < 0$ and instability when $f'(a) > 0$.
- b) **NDC** *Non-hyperbolic Derivative Caveat.* For $y' = y^3$ and $y' = -y^3$, determine the sign of y' on either side of the equilibrium $a = 0$ and prove that 0 is unstable in the first case and asymptotically stable in the second, although $f'(0) = 0$ in both.

Indication.— For part (a), continuity of f' gives a punctured interval on which $f(y)$ points towards a when $f'(a) < 0$ and away from a when $f'(a) > 0$. Use SAUS to prevent crossing. A bounded monotone solution has a limit; if that limit were a nearby point different from a , continuity would make its derivative stay bounded away from zero.

40) Let $a_0, b_0, \lambda \in (0, \infty)$ satisfy $a_0 < b_0$. In a dilute solution, let $a, b, c: [0, \infty) \rightarrow [0, \infty)$ be continuous and differentiable for $t > 0$, denoting the amounts of NaOH, HCl, and each reaction product. Suppose $a(0) = a_0$, $b(0) = b_0$, and $c(0) = 0$. Assume the reaction rate is $\lambda a(t)b(t)$ for $t > 0$.

- a) **MBM** *Mass Balance Model.* Use the reaction stoichiometry to obtain $a = a_0 - c$, $b = b_0 - c$, and

$$c' = \lambda(a_0 - c)(b_0 - c).$$

- b) **ECSA** *Equilibria and Chemical Stability Analysis.* Find the equilibria $c = a_0$ and $c = b_0$. Use the sign of the right-hand side, or Question 39, to determine which is asymptotically stable and which is unstable.
- c) **SCNF** *Separable Chemical Neutralization Formula.* Separate variables, use partial fractions, impose $c(0) = 0$, and derive

$$c(t) = \frac{a_0 b_0 (1 - e^{-(b_0 - a_0)\lambda t})}{b_0 - a_0 e^{-(b_0 - a_0)\lambda t}} \quad (t \geq 0).$$

- d) **PDR** *Physical Domain Restriction.* Prove that $0 \leq c(t) < a_0$ for every finite $t \geq 0$ and that $c(t) \rightarrow a_0$ as $t \rightarrow \infty$. Deduce that the branch with $c > a_0$ makes one reactant amount negative and is therefore excluded.

C. — Continuous and discrete logistic growth

Logistic equation. Let $r, Y \in (0, \infty)$. The *logistic equation* for a population $y: [0, \infty) \rightarrow [0, \infty)$ is

$$y' = ry \left(1 - \frac{y}{Y}\right),$$

where r is the net low-density growth rate and Y is the *carrying capacity*. In the model of the notes, $r = \alpha - \beta > 0$ for per-capita birth and death rates $\alpha, \beta \in [0, \infty)$, and competition adds a death term γy^2 with $\gamma > 0$, giving $Y = r/\gamma$.

41)

- a) **ELG** *Exponential Low-density Growth.* Let $\rho \in \mathbb{R}$ and $y_0 \in [0, \infty)$. Omit competition and solve $y' = \rho y$ with $y(0) = y_0$. Show that y is strictly increasing when $\rho > 0$ and strictly decreasing when $\rho < 0$.
- b) **CPC** *Competition Produces Carrying capacity.* Starting from $(\alpha - \beta)y - \gamma y^2$, factor the right-hand side into logistic form and determine Y .

- c) **LESC** *Logistic Equilibria and Stability Classification.* Find the equilibria 0 and Y , compute the derivative of the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(y) = ry(1 - y/Y)$ at each, and determine their stability using Question 39.
- d) **LES** *Logistic Explicit Solution.* For $y_0 \in (0, \infty)$, separate variables and prove

$$y(t) = \frac{Y}{1 + (Y/y_0 - 1)e^{-rt}} \quad (t \geq 0).$$

Verify its low-density approximation and its limit as $t \rightarrow \infty$.

Discrete dynamical systems. Let $S \subseteq \mathbb{R}$ and let $F: S \rightarrow S$. A *difference equation*, or *discrete dynamical system*, is a sequence $(x_n)_{n \in \mathbb{N}_0}$ satisfying $x_{n+1} = F(x_n)$ for every $n \in \mathbb{N}_0$.

Fixed points of maps. A *fixed point* of $F: S \rightarrow S$ is a point $x^* \in S$ with $F(x^*) = x^*$.

Stable fixed points. A fixed point x^* is *stable* if, for every $\epsilon > 0$, there exists $\delta > 0$ such that $|x_0 - x^*| < \delta$ implies $|x_n - x^*| < \epsilon$ for every $n \in \mathbb{N}_0$. It is *unstable* if it is not stable.

Locally attracting fixed points. A fixed point x^* is *locally attracting* if all sufficiently close initial values produce $x_n \rightarrow x^*$ as $n \rightarrow \infty$.

The logistic map. For $\lambda > 0$, the *logistic map* is the function $F_\lambda: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$F_\lambda(x) = \lambda x(1 - x), \quad x_{n+1} = F_\lambda(x_n).$$

Cobweb diagrams. A *cobweb diagram* alternates vertical segments to the graph of F_λ and horizontal segments to the diagonal $x_{n+1} = x_n$, thereby displaying successive iterates.

42)

- a) **FELM** *Forward Euler Logistic Map.* Let $\Delta t > 0$ and $y_0 \in [0, \infty)$. Apply a forward difference of step Δt to the continuous logistic equation to obtain a recurrence for a real sequence $(y_n)_{n \in \mathbb{N}_0}$ with initial value y_0 . With

$$\lambda = 1 + r\Delta t, \quad x_n = \frac{r\Delta t}{1 + r\Delta t} \frac{y_n}{Y},$$

derive $x_{n+1} = \lambda x_n(1 - x_n)$.

- b) **DFP** *Discrete Fixed Points.* Solve $F_\lambda(x) = x$ and obtain $x = 0$ and, when $\lambda \neq 0$, $x = 1 - 1/\lambda$.
- c) **DLSC** *Discrete Linear Stability Criterion.* Let $J \subseteq \mathbb{R}$ be an open interval, let $F: J \rightarrow J$ be twice differentiable near a fixed point $x^* \in J$. Put $x_n = x^* + \varepsilon_n$ and show $\varepsilon_{n+1} = F'(x^*)\varepsilon_n + O(\varepsilon_n^2)$. Deduce local attraction when $|F'(x^*)| < 1$ and instability when $|F'(x^*)| > 1$.
- d) **PRC** *Parameter Regime Classification.* Compute F'_λ at both fixed points. Show that 0 is attracting for $0 < \lambda < 1$, while $1 - 1/\lambda$ is attracting for $1 < \lambda < 3$. Show also that the displacement $\varepsilon_n = x_n - x^*$ keeps a constant sign when $1 < \lambda < 2$ and alternates in sign when $2 < \lambda < 3$.

Indication. — For part (c), when $|F'(x^*)| < 1$, use continuity of F' and the mean-value theorem to find a neighbourhood on which F contracts distances from x^* . When $|F'(x^*)| > 1$, find a neighbourhood on which each non-zero displacement expands until it leaves that neighbourhood.

Periodic orbits. Retain a set $S \subseteq \mathbb{R}$ and a function $F: S \rightarrow S$. A *periodic orbit of least period* $k \in \mathbb{N}$ is a list of distinct points $x_0, \dots, x_{k-1} \in S$ such that $F(x_j) = x_{j+1}$ for every $j \in \mathbb{N}_0$ with $0 \leq j < k-1$, $F(x_{k-1}) = x_0$, and no smaller positive period has this property. It is also called a limit cycle in the terminology of the original notes.

43) Let $F_\lambda(x) = \lambda x(1 - x)$ with $\lambda > 0$.

- a) **TPE** *Two-cycle Polynomial Extraction.* Compute $F_\lambda(F_\lambda(x)) - x$, divide out the factors corresponding to the two fixed points, and show that the remaining quadratic has two distinct real roots precisely when $\lambda > 3$.

Indication.— After removing x and $x - (1 - 1/\lambda)$, the quadratic may be written $\lambda^2 x^2 - \lambda(\lambda + 1)x + (\lambda + 1) = 0$.

- b) **TCS** *Two-Cycle Stability.* If x_1, x_2 are the two roots from part (a), show that the derivative of the second iterate at either point is $F'_\lambda(x_1)F'_\lambda(x_2)$. Compute its absolute value and prove that the two-cycle is attracting exactly when $3 < \lambda < 1 + \sqrt{6}$.

Admitted result. **LMPDB** *Logistic Map Period-Doubling Behaviour.* As λ increases beyond $1 + \sqrt{6}$, the logistic map's attracting two-cycle loses stability and an attracting four-cycle appears. Further increases produce a cascade of period doublings, including cycles of periods 8, 16, ..., followed by parameter ranges with chaotic dynamics, in which the long-term behaviour is not eventually periodic and depends sensitively on the initial value. The earlier fixed point and periodic orbits continue to exist after losing stability.

End of Part IV

Part V. — Homogeneous second-order equations and phase space

Second-order equations arise when a model contains acceleration. This part classifies constant-coefficient homogeneous solutions and gives a geometric meaning to the two initial data required by an equation of order two.

Second-order linear equations. Let $I \subseteq \mathbb{R}$ be an interval, let $a, b, c \in \mathbb{R}$ with $a \neq 0$, and let $f: I \rightarrow \mathbb{R}$. A *second-order linear ordinary differential equation with constant coefficients* has the form

$$ay''(x) + by'(x) + cy(x) = f(x) \quad (x \in I),$$

for an unknown twice differentiable function $y: I \rightarrow \mathbb{R}$.

Homogeneous second-order equations. The equation above is *homogeneous* when f is the zero function. For an inhomogeneous equation, linearity gives the same two-step strategy as in Part III: find the general complementary function y_c and one particular integral y_p , then write $y = y_c + y_p$, where $y_c, y_p: I \rightarrow \mathbb{R}$.

Admitted result. LIVPEU Linear Initial-Value Problem Existence and Uniqueness. Let $I \subseteq \mathbb{R}$ be an open interval, let $p, q, r: I \rightarrow \mathbb{R}$ be continuous, let $x_0 \in I$, and let $u_0, v_0 \in \mathbb{R}$. There exists a unique twice continuously differentiable function $y: I \rightarrow \mathbb{R}$ satisfying

$$y'' + p(x)y' + q(x)y = r(x) \quad (x \in I), \quad y(x_0) = u_0, \quad y'(x_0) = v_0.$$

More generally, let $n \in \mathbb{N}$, let $I \subseteq \mathbb{R}$ be an open interval, let $a_0, \dots, a_n, r: I \rightarrow \mathbb{R}$ be continuous, and suppose $a_n(x) \neq 0$ for every $x \in I$. For every $x_0 \in I$ and every $(u_0, \dots, u_{n-1}) \in \mathbb{R}^n$, there exists a unique n -times continuously differentiable function $y: I \rightarrow \mathbb{R}$ satisfying

$$\sum_{j=0}^n a_j(x)y^{(j)}(x) = r(x) \quad (x \in I), \quad y^{(j)}(x_0) = u_j \quad (j \in \mathbb{N}_0, 0 \leq j < n).$$

A. — Characteristic roots

44) Let $a, b, c \in \mathbb{R}$ with $a \neq 0$ and seek complex-valued solutions of $ay'' + by' + cy = 0$ on $\mathbb{R}$.

- a) **CEFE Characteristic Equation From an Exponential.** Substitute $y(x) = e^{\lambda x}$ with $\lambda \in \mathbb{C}$ and prove that it is a solution if and only if

$$a\lambda^2 + b\lambda + c = 0.$$

This polynomial equation is called the *characteristic equation*.

- b) **LIDS Linear Independence for Distinct Solutions.** Let $\lambda_1, \lambda_2 \in \mathbb{C}$ be distinct characteristic roots. If $A, B \in \mathbb{C}$ satisfy $Ae^{\lambda_1 x} + Be^{\lambda_2 x} = 0$ for every $x \in \mathbb{R}$, evaluate the identity and its derivative at $x = 0$ and prove $A = B = 0$.
- c) **DRGS Distinct-Root General Solution.** For arbitrary $u_0, v_0 \in \mathbb{C}$, impose the complex initial data $y(0) = u_0$ and $y'(0) = v_0$, and solve uniquely for $A, B \in \mathbb{C}$ in $y = Ae^{\lambda_1 x} + Be^{\lambda_2 x}$. Use LIVPEU, applied to real and imaginary parts, to deduce that this is the general complex-valued solution.
- d) **DRWE Distinct Real-root Worked Example.** Solve $y'' - 5y' + 6y = 0$ on $\mathbb{R}$ and obtain $y = Ae^{2x} + Be^{3x}$, where $A, B \in \mathbb{R}$ for real-valued solutions.

45)

- a) **CSHM** *Complex Solutions of Harmonic Motion.* Solve the characteristic equation for $y'' + 4y = 0$ and obtain the complex basis e^{2ix}, e^{-2ix} .
- b) **RBCT** *Real Basis through Complex Transformation.* Use $e^{i\theta} = \cos \theta + i \sin \theta$ to show that every real-valued solution can be written uniquely as

$$y(x) = \alpha \cos(2x) + \beta \sin(2x)$$

with $\alpha, \beta \in \mathbb{R}$. Prove that y is real-valued precisely when $B = \overline{A}$ and that then $\alpha = A + B$ and $\beta = i(A - B)$.

- c) **DRSS** *Detuning Reveals a Second Solution.* For $\varepsilon \in \mathbb{R} \setminus \{0\}$, solve

$$y'' - 4y' + (4 - \varepsilon^2)y = 0.$$

Expand $e^{\pm \varepsilon x}$ at $\varepsilon = 0$ and show how the two combinations

$$\frac{e^{(2+\varepsilon)x} + e^{(2-\varepsilon)x}}{2}, \quad \frac{e^{(2+\varepsilon)x} - e^{(2-\varepsilon)x}}{2\varepsilon}$$

tend respectively to e^{2x} and xe^{2x} .

- d) **RRGS** *Repeated-Root General Solution.* Let $a\lambda^2 + b\lambda + c$ have a repeated root $\lambda_0 \in \mathbb{R}$. Verify directly that $e^{\lambda_0 x}$ and $xe^{\lambda_0 x}$ solve the equation. Show that their initial state vectors at $x = 0$ are $(1, \lambda_0)$ and $(0, 1)$, and hence that their linear combinations realize arbitrary data in $\mathbb{R}^2$. Use LIVPEU to prove that the general real solution is

$$y(x) = (A + Bx)e^{\lambda_0 x}, \quad A, B \in \mathbb{R}.$$

Reduction of order. Given one non-zero solution $y_1: I \rightarrow \mathbb{R}$ of a homogeneous second-order linear equation, *reduction of order* seeks another solution in the form $y_2 = vy_1$, where $v: I \rightarrow \mathbb{R}$ is an unknown twice differentiable function.

46) For $y'' - 4y' + 4y = 0$ on $\mathbb{R}$, take $y_1(x) = e^{2x}$, let $v: \mathbb{R} \rightarrow \mathbb{R}$ be twice differentiable, and set $y_2(x) = v(x)e^{2x}$ for every $x \in \mathbb{R}$.

- a) **ROA** *Reduction of Order Ansatz.* Compute y_2' and y_2'' in terms of v, v' , and v'' .
- b) **REEC** *Repeated Equation Exact Cancellation.* Substitute these expressions and prove that every term except $v''(x)e^{2x}$ cancels.
- c) **SVLV** *Second solution Via a Linear Variation.* Solve $v'' = 0$, discard the component proportional to y_1 , and recover the second solution $y_2(x) = xe^{2x}$.

B. — Phase space and the Wronskian

Higher-order linear equations. For a positive integer $n \in \mathbb{N}$, let $I \subseteq \mathbb{R}$ be an interval and consider

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = f(x),$$

where $a_0, \dots, a_n, f: I \rightarrow \mathbb{R}$ and $a_n(x) \neq 0$ for every $x \in I$, for an unknown n -times differentiable function $y: I \rightarrow \mathbb{R}$.

State vectors. The *state vector* or *solution vector* of the equation above is the function $\mathbf{Y}: I \rightarrow \mathbb{R}^n$ defined by

$$\mathbf{Y}(x) = (y(x), y'(x), \dots, y^{(n-1)}(x)).$$

Phase space. The ambient space $\mathbb{R}^n$ of possible states is the *phase space*; a solution traces a phase-space trajectory.

47)

- a) **SFOS** *State First-Order System*. Solve the displayed differential equation algebraically for $y^{(n)}$ and show that it determines the derivative of every component of $\mathbf{Y}$ from x and $\mathbf{Y}(x)$.
- b) **PID** *Phase Initial Data*. Assume that I is open and that $a_0, \dots, a_n$ and f are continuous on I , and let $x_0 \in I$. Show that an initial state $\mathbf{Y}(x_0) = (u_0, \dots, u_{n-1}) \in \mathbb{R}^n$ is exactly the data $y(x_0) = u_0, \dots, y^{(n-1)}(x_0) = u_{n-1}$, and use LIVPEU to prove that there is exactly one solution with this initial state.
- c) **SHPT** *Simple Harmonic Phase Trajectory*. For $y'' + 4y = 0$ with $y(0) = 1$ and $y'(0) = 0$, find y and prove that

$$\mathbf{Y}(x) = (\cos 2x, -2 \sin 2x)$$

traces the ellipse $y^2 + (y'/2)^2 = 1$ clockwise as x increases from 0.

- d) **SIBT** *Second Independent Basis Trajectory*. Repeat part (c) for $y(0) = 0$ and $y'(0) = 2$. Show that the two initial state vectors $(1, 0)$ and $(0, 2)$ form a basis of $\mathbb{R}^2$ and that the corresponding state vectors remain independent for every $x \in \mathbb{R}$.

Wronskian. Let $I \subseteq \mathbb{R}$ be an interval. For differentiable functions $y_1, y_2: I \rightarrow \mathbb{R}$, their *Wronskian* is the function $W: I \rightarrow \mathbb{R}$ defined by the 2×2 determinant

$$W(x) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix} = y_1(x)y_2'(x) - y_2(x)y_1'(x).$$

Linear independence of functions. Functions $y_1, y_2: I \rightarrow \mathbb{R}$ are *linearly independent* if the identity $Ay_1 + By_2 = 0$ on I , for constants $A, B \in \mathbb{R}$, implies $A = B = 0$.

48) Let $I \subseteq \mathbb{R}$ be an open interval, let $p, q: I \rightarrow \mathbb{R}$ be continuous, and let $y_1, y_2: I \rightarrow \mathbb{R}$ solve $y'' + p(x)y' + q(x)y = 0$ on I .

- a) **WIC** *Wronskian Independence Certificate*. Prove that $W(x_0) \neq 0$ at some $x_0 \in I$ implies linear independence. Conversely, if $W(x_0) = 0$, choose constants A, B , not both zero, so that $Ay_1 + By_2$ and its derivative vanish at x_0 , and use LIVPEU to prove dependence.
- b) **WSH** *Wronskian for Simple Harmonics*. Compute the Wronskian of $\cos 2x$ and $\sin 2x$ and obtain $W(x) = 2$ for every $x \in \mathbb{R}$.
- c) **WRR** *Wronskian for a Repeated Root*. Compute the Wronskian of e^{2x} and xe^{2x} and obtain $W(x) = e^{4x}$ for every $x \in \mathbb{R}$.

49) Let $I \subseteq \mathbb{R}$ be an open interval, let $p, q: I \rightarrow \mathbb{R}$ be continuous, and let $y_1, y_2: I \rightarrow \mathbb{R}$ solve $y'' + p(x)y' + q(x)y = 0$ on I .

- a) **SWI** *Subtracted Wronskian Identity*. Multiply the equation for y_1 by y_2 , multiply that for y_2 by y_1 , subtract, and express the result in terms of W and W' .
- b) **AWE** *Abel Wronskian Equation*. Prove $W' + pW = 0$ and apply Question 32 to obtain

$$W(x) = W(x_0) \exp \left(- \int_{x_0}^x p(s) \, ds \right)$$

for all $x, x_0 \in I$.

- c) **ZONZ** *Zero Or Never Zero*. Deduce Abel's theorem: the Wronskian is either zero throughout I or non-zero at every point of I . Combine this with Question 48 to prove that y_1 and y_2 are linearly independent if and only if they are linearly independent at every point of I .

Part IX will define the coefficient matrix, its trace, and the higher-order Wronskian before formulating the corresponding determinant identity for systems of first-order equations.

End of Part V

Part VI. — Forced and equidimensional equations, recurrences, damping, and impulses

This part completes the constant-coefficient theory and compares differential equations with two related structures: equidimensional equations and linear recurrences. It ends with physical forcing over long and very short time scales.

A. — Particular integrals and resonance

Undetermined coefficients. For constants $a, b, c \in \mathbb{R}$ with $a \neq 0$, write $L[y] = ay'' + by' + cy$. The *method of undetermined coefficients* chooses a trial particular integral whose algebraic form resembles a simple forcing term and determines its coefficients by substitution. Linearity allows separate particular integrals to be added. Useful trial spaces are

$$Ae^{mx}, \quad A \sin(kx) + B \cos(kx), \quad A_n x^n + \cdots + A_1 x + A_0$$

for forcing terms e^{mx} , $\sin(kx)$ or $\cos(kx)$, and a polynomial of degree n , provided none of the trial functions already belongs to the complementary solution space; resonant forcing is treated below.

50)

- a) **PEFC** *Polynomial and Exponential Forcing Computation.* For

$$y'' - 5y' + 6y = 2x + e^{4x},$$

use $y_p = ax + b + ce^{4x}$ and prove $a = 1/3$, $b = 5/18$, and $c = 1/2$.

- b) **CSFS** *Complementary Superposition Forms the Solution.* Determine the complementary function and prove that the general solution is

$$y(x) = Ae^{2x} + Be^{3x} + \frac{1}{3}x + \frac{5}{18} + \frac{1}{2}e^{4x}, \quad A, B \in \mathbb{R}.$$

Resonance. For a linear equation $L[y] = f$, *resonance* occurs when the forcing f is itself a complementary solution, so the usual same-form trial is annihilated by L .

Beating. *Beating* is slow amplitude modulation caused by forcing at a frequency close to the natural frequency ω_0 of the undamped oscillator $y'' + \omega_0^2 y = 0$.

51) Let $\omega_0 \in (0, \infty)$.

- a) **SGRF** *Standard Guess Resonance Failure.* For $y'' + \omega_0^2 y = \sin(\omega_0 t)$, prove that every trial $C \sin(\omega_0 t) + D \cos(\omega_0 t)$ is annihilated by the left-hand operator and hence cannot be a particular integral.

- b) **DNPI** *Detuned Nonresonant Particular Integral.* Let $\omega \in (0, \infty) \setminus \{\omega_0\}$. Verify that

$$y_p(t) = \frac{\sin(\omega t) - \sin(\omega_0 t)}{\omega_0^2 - \omega^2}$$

solves $y'' + \omega_0^2 y = \sin(\omega t)$, and prove that adding any solution of the homogeneous equation to y_p leaves the forcing unchanged.

- c) **BLS** *Beating Length Scales.* Put $\Delta\omega = \omega_0 - \omega$ and use a product-to-sum identity to express y_p as a fast cosine times a slow sine envelope. Determine the respective time scales $O(1/\omega_0)$ and $O(1/|\Delta\omega|)$ when $|\Delta\omega| \ll \omega_0$.

- d) **RPL** *Resonant Particular Limit.* For each fixed $t \in \mathbb{R}$, take $\omega \rightarrow \omega_0$, using $\sin z/z \rightarrow 1$, and prove

$$y_p(t) \longrightarrow -\frac{t}{2\omega_0} \cos(\omega_0 t).$$

Verify this limiting function directly in the resonant equation.

- e) **GRM** *General Resonance Multiplier*. Let $a, b, c \in \mathbb{R}$ with $a \neq 0$, and let $P: \mathbb{C} \rightarrow \mathbb{C}$ be the polynomial with real coefficients

$$P(z) = az^2 + bz + c \quad (z \in \mathbb{C}).$$

Suppose that $\lambda \in \mathbb{C}$ is a simple root of P . For $L[y] = ay'' + by' + cy$, verify

$$L[Axe^{\lambda x}] = A(2a\lambda + b)e^{\lambda x},$$

and choose $A \in \mathbb{C}$ so this expression equals $e^{\lambda x}$. Determine the analogous real sine-cosine trial multiplied by x when the simple characteristic roots are $\pm i\omega_0$ with $\omega_0 > 0$.

Variation of parameters. Let $I \subseteq \mathbb{R}$ be an open interval, let $p, q, f: I \rightarrow \mathbb{R}$ be continuous, and let $y_1, y_2: I \rightarrow \mathbb{R}$ be linearly independent solutions of $y'' + py' + qy = 0$. Their Wronskian W never vanishes by Question 48. Variation of parameters seeks a particular solution of $y'' + py' + qy = f$ in the form $y_p = uy_1 + vy_2$, where $u, v: I \rightarrow \mathbb{R}$ are continuously differentiable coefficient functions.

52)

- a) **VPA** *Variation Parameter Ansatz*. Differentiate $y_p = uy_1 + vy_2$ and impose the auxiliary condition $u'y_1 + v'y_2 = 0$ to obtain $y'_p = uy'_1 + vy'_2$.
- b) **ADC** *Auxiliary Derivative Constraint*. Differentiate once more, substitute into the inhomogeneous equation, and use the two homogeneous equations to prove $u'y'_1 + v'y'_2 = f$.
- c) **WLS** *Wronskian Linear System*. Write the two equations for (u', v') as

$$\begin{pmatrix} y_1 & y_2 \\ y'_1 & y'_2 \end{pmatrix} \begin{pmatrix} u' \\ v' \end{pmatrix} = \begin{pmatrix} 0 \\ f \end{pmatrix}$$

and solve them to obtain $u' = -y_2 f / W$ and $v' = y_1 f / W$.

- d) **VPIF** *Variation Parameters Integral Formula*. Fix $x_0 \in I$, integrate the preceding derivatives, and prove that

$$y_p(x) = -y_1(x) \int_{x_0}^x \frac{y_2(s)f(s)}{W(s)} ds + y_2(x) \int_{x_0}^x \frac{y_1(s)f(s)}{W(s)} ds$$

is a particular solution. Show that changing either integration constant adds a complementary solution to y_p .

- 53) For $y'' + 4y = \sin(2x)$, take $y_1(x) = \sin(2x)$ and $y_2(x) = \cos(2x)$.

- a) **WSEC** *Wronskian Sign and Elementary Coefficients*. Compute $W = -2$ and derive

$$u' = \frac{1}{4} \sin(4x), \quad v' = \frac{1}{4} (\cos(4x) - 1).$$

- b) **VPIE** *Variation Parameter Integration Example*. Integrate, substitute in $uy_1 + vy_2$, and simplify to a particular integral differing from $-(x/4) \cos(2x)$ by at most a complementary function.
- c) **DVA** *Detuning and Variation Agree*. Verify directly that $-(x/4) \cos(2x)$ satisfies $y'' + 4y = \sin(2x)$, consistently with the resonant limit in Question 51.

B. — Equidimensional equations and recurrences

Equidimensional equation. On $(0, \infty)$, a second-order *equidimensional*, *Euler*, or *Cauchy–Euler* equation has the form

$$ax^2y'' + bxy' + cy = f(x),$$

where $a, b, c \in \mathbb{R}$, $a \neq 0$, and $f: (0, \infty) \rightarrow \mathbb{R}$ is continuous, for an unknown twice differentiable function $y: (0, \infty) \rightarrow \mathbb{R}$. The powers of x make every term have the dimensions of y : if $[x] = T$ and $[y] = L$, then $[x^2 y''] = [xy'] = [y] = L$. The equation is *homogeneous* when f is the zero function. Some texts use “homogeneous equation” as a synonym for “equidimensional equation”; here that phrase always means that the forcing is zero.

54)

- a) **IEPP** *Indicial Equation from a Power Profile*. Substitute $y = x^k$ with $k \in \mathbb{R}$ in the homogeneous equation and derive the *indicial equation*

$$ak(k-1) + bk + c = 0.$$

- b) **DRCF** *Distinct-Root Complementary Form*. If the indicial roots $k_1, k_2 \in \mathbb{R}$ are distinct, verify that $Ax^{k_1} + Bx^{k_2}$ is a solution for all $A, B \in \mathbb{R}$.
- c) **LTCC** *Logarithmic Transformation to Constant Coefficients*. Put $z = \log x$ and let $Y: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $Y(z) = y(e^z)$. Prove

$$xy' = Y_z, \quad x^2 y'' = Y_{zz} - Y_z,$$

and transform the equation into

$$aY_{zz} + (b-a)Y_z + cY = f(e^z).$$

Use LIVPEU to justify that the two functions in part (b) span the homogeneous solution space.

- d) **RRL** *Repeated Root Logarithm*. If the indicial equation has a repeated real root k , use the transformed constant-coefficient equation to prove

$$y_c(x) = (A + B \log x)x^k.$$

- e) **RFR** *Resonant Forcing Response*. Suppose k_1 is a simple indicial root and $f(x)$ is proportional to x^{k_1} . Transform to z and show that a particular trial of the form $Cx^{k_1} \log x$ leads to a solvable equation for C . Determine the extra factor required when the indicial root is repeated.

Second-order linear difference equations. Let $a, b, c \in \mathbb{R}$ with $a \neq 0$, and let $f: \mathbb{N}_0 \rightarrow \mathbb{R}$. A second-order linear difference equation with constant coefficients is

$$ay_{n+2} + by_{n+1} + cy_n = f_n \quad (n \in \mathbb{N}_0)$$

for an unknown sequence $y: \mathbb{N}_0 \rightarrow \mathbb{R}$, where $y_n = y(n)$ and $f_n = f(n)$. It is homogeneous when $f_n = 0$ for every $n \in \mathbb{N}_0$.

Shift operator. The *shift operator* is $S[y]_n = y_{n+1}$.

55)

- a) **SEGP** *Shift Eigenfunction as a Geometric Progression*. For $k \in \mathbb{R}$, let the sequence $g = (g_n)_{n \in \mathbb{N}_0}$ be defined by $g_n = k^n$, using $0^0 = 1$. Show that $S[g]_n = kg_n$ for every $n \in \mathbb{N}_0$.
- b) **DCEC** *Difference Characteristic Equation Calculation*. Substitute $y_n = k^n$ and derive the *characteristic equation* $ak^2 + bk + c = 0$, whose coefficients $a, b, c \in \mathbb{R}$ satisfy $a \neq 0$. If its two roots k_1, k_2 are distinct and non-zero, prove that $y_n = Ak_1^n + Bk_2^n$ realizes arbitrary initial data y_0, y_1 and hence is the general homogeneous solution.
- c) **RRCF** *Repeated root Recurrence Complementary Form*. If the characteristic equation has a repeated non-zero root k , verify directly that k^n and nk^n are solutions and deduce $y_n = (A + Bn)k^n$.

- d) **DZRH** *Degenerate Zero-Root Handling.* If $c = 0$, factor the recurrence operator and solve the recurrence when the zero characteristic root is simple and when it is repeated. Show in particular that the formula in part (c) does not apply when $k = 0$.
- e) **PIG** *Particular Inspection Guesses.* Let $r \in \mathbb{R} \setminus \{0\}$ and $p \in \mathbb{N}_0$. Give suitable trial forms for forcing r^n when r is not a characteristic root and for polynomial forcing of degree p when 1 is not a characteristic root, and show by substitution that the coefficients are determined. When r is a simple characteristic root, show that the trial Ar^n fails and that Anr^n succeeds; determine the additional factor required for a repeated root.
- 56) Let $(F_n)_{n \in \mathbb{N}_0}$ satisfy $F_{n+2} = F_{n+1} + F_n$ for every $n \in \mathbb{N}_0$, with $F_0 = F_1 = 1$. Put $\varphi_1 = (1 + \sqrt{5})/2$ and $\varphi_2 = (1 - \sqrt{5})/2$.

- a) **FCR** *Fibonacci Characteristic Roots.* Derive the characteristic equation $k^2 - k - 1 = 0$ and its two roots φ_1, φ_2 .
- b) **BIC** *Binet Initial Coefficients.* Determine $A, B \in \mathbb{R}$ in $F_n = A\varphi_1^n + B\varphi_2^n$ and prove

$$F_n = \frac{\varphi_1^{n+1} - \varphi_2^{n+1}}{\sqrt{5}} \quad (n \in \mathbb{N}_0).$$

Use $\varphi_1\varphi_2 = -1$ to write $\varphi_2 = -1/\varphi_1$ and obtain the equivalent form

$$F_n = \frac{\varphi_1^{n+1} - (-1/\varphi_1)^{n+1}}{\sqrt{5}} \quad (n \in \mathbb{N}_0).$$

C. — Damping and sustained forcing

Damped oscillator model. Let $M, k \in (0, \infty)$ be a mass and spring constant, let $l \in [0, \infty)$ be a damping coefficient, let $F: [0, \infty) \rightarrow \mathbb{R}$ be a continuous applied force, and let $x: [0, \infty) \rightarrow \mathbb{R}$ be a twice continuously differentiable vertical displacement. The restoring and damping forces are $-kx$ and $-l\dot{x}$.

57)

- a) **NFM** *Newton Force Model.* Apply Newton's second law to obtain

$$M\ddot{x} + l\dot{x} + kx = F(t).$$

- b) **NDS** *Nondimensional Derivative Scaling.* Put

$$\begin{aligned} \tau &= t\sqrt{k/M}, & \kappa &= \frac{l}{2\sqrt{kM}}, \\ X(\tau) &= x\left(\tau\sqrt{M/k}\right), & f(\tau) &= \frac{F\left(\tau\sqrt{M/k}\right)}{k}, \end{aligned}$$

where $X: [0, \infty) \rightarrow \mathbb{R}$ and $f: [0, \infty) \rightarrow \mathbb{R}$. Transform derivatives carefully and derive $X'' + 2\kappa X' + X = f(\tau)$. For the rest of Part C, relabel X as x and let dots denote differentiation with respect to τ .

- c) **FRC** *Free Response Characteristic equation.* For $f = 0$, derive the roots $-\kappa \pm \sqrt{\kappa^2 - 1}$.
- 58) Assume $\kappa > 0$.

- a) **UOD** *Underdamped Oscillatory Decay.* If $0 < \kappa < 1$, prove

$$x(\tau) = e^{-\kappa\tau} \left(A \sin(\sqrt{1 - \kappa^2} \tau) + B \cos(\sqrt{1 - \kappa^2} \tau) \right).$$

Here $A, B \in \mathbb{R}$. Determine the oscillation period and the envelope decay time.

- b) **CDR** *Critically Damped Response*. If $\kappa = 1$, derive $x(\tau) = (A + B\tau)e^{-\tau}$, where $A, B \in \mathbb{R}$, and determine its dimensionless and dimensional rise and decay scales.
- c) **ODTS** *Overdamped Dual Time Scales*. If $\kappa > 1$, put $\lambda_1 = \kappa - \sqrt{\kappa^2 - 1}$ and $\lambda_2 = \kappa + \sqrt{\kappa^2 - 1}$. Prove $x = Ae^{-\lambda_1\tau} + Be^{-\lambda_2\tau}$, where $A, B \in \mathbb{R}$, and determine the slow decay and fast rise scales.
- d) **TGPD** *Transient Growth Precedes Decay*. Give choices of A, B in either the critical or overdamped case for which $|x|$ initially increases even though $x(\tau) \rightarrow 0$ as $\tau \rightarrow \infty$.

59) Let $\kappa \in (0, \infty)$ and consider $\ddot{x} + 2\kappa\dot{x} + x = \sin \tau$, where dots denote differentiation with respect to τ .

- a) **SFPI** *Sinusoidal Forcing Particular Integral*. Substitute $x_p = C \sin \tau + D \cos \tau$ and prove $x_p = -(2\kappa)^{-1} \cos \tau$.
- b) **TAA** *Transient Approaches Asymptotic response*. Add the appropriate complementary function from Question 58. Prove that it tends to zero and hence

$$x(\tau) = -\frac{1}{2\kappa} \cos \tau + o(1) \quad (\tau \rightarrow \infty).$$

Determine the phase shift between $\sin \tau$ and the long-time response $-(2\kappa)^{-1} \cos \tau$.

D. — Impulses and step inputs

Impulse. The *impulse* of a force F acting near time T is its time integral.

Nascent delta families. A *nascent delta family* is a family $D_\varepsilon: \mathbb{R} \rightarrow [0, \infty)$, indexed by $\varepsilon > 0$, such that $D_\varepsilon(t) \rightarrow 0$ for every $t \neq 0$ and $\int_{-\infty}^{\infty} D_\varepsilon(t) dt \rightarrow 1$ as $\varepsilon \rightarrow 0^+$.

Admitted result. **GIN** *Gaussian Integral Normalization*. The improper integral

$$\int_{-\infty}^{\infty} e^{-u^2} du$$

exists and equals $\sqrt{\pi}$.

60)

- a) **MJI** *Momentum Jump from an Impulse*. Let $T \in \mathbb{R}$ and $\varepsilon > 0$. Suppose a ball of mass $m > 0$ has a twice continuously differentiable position $x: [T - \varepsilon, T + \varepsilon] \rightarrow \mathbb{R}$ and is subject to a continuous contact force $F: [T - \varepsilon, T + \varepsilon] \rightarrow \mathbb{R}$. If $m\ddot{x} = F(t) - mg$ on this interval, where $g > 0$, integrate and prove

$$m\dot{x}(T + \varepsilon) - m\dot{x}(T - \varepsilon) = I_\varepsilon - 2\varepsilon mg,$$

where $I_\varepsilon = \int_{T-\varepsilon}^{T+\varepsilon} F(t) dt$.

- b) **GND** *Gaussian Nascent Delta*. For $\varepsilon > 0$, let $D_\varepsilon: \mathbb{R} \rightarrow [0, \infty)$ be defined by

$$D_\varepsilon(t) = \frac{1}{\varepsilon\sqrt{\pi}} e^{-t^2/\varepsilon^2}.$$

Using the substitution $u = t/\varepsilon$ and GIN, prove that its integral is 1, its maximum height is $D_\varepsilon(0) = 1/(\varepsilon\sqrt{\pi})$, and $D_\varepsilon(\pm\varepsilon) = e^{-1}D_\varepsilon(0)$. Deduce that its height is $O(1/\varepsilon)$ and that the width over which it decays by a factor e is $O(\varepsilon)$, and prove that it tends pointwise to 0 away from the origin.

Admitted result. **DSP** *Dirac Sifting Property*. The Dirac delta δ is the generalized object defined by its action under integration. If $a, b, c \in \mathbb{R}$ satisfy $a < b$ and $g: [a, b] \rightarrow \mathbb{R}$ is continuous, then

$$\int_a^b g(x)\delta(x - c) dx = \begin{cases} g(c), & c \in (a, b), \\ 0, & c \notin [a, b]. \end{cases}$$

At an endpoint $c = a$ or $c = b$, a separate limiting convention is required. The identity represents the corresponding limit against a standard nascent delta family and is used only inside integrals.

If the impulses I_ε in MJI tend to $I \in \mathbb{R}$ as $\varepsilon \rightarrow 0^+$, the corresponding zero-duration contact model is $m\ddot{x} = -mg + I\delta(t - T)$ in the generalized sense of DSP.

61) Consider, in the generalized sense specified by DSP,

$$y'' - y = 3\delta(x - \pi/2), \quad y(0) = y(\pi) = 0,$$

and require y to be continuous on $[0, \pi]$ and twice differentiable away from $x = \pi/2$.

a) **PRHS** *Piecewise Regional Homogeneous Solutions.* Solve the homogeneous equation on each side and use the endpoint conditions to write

$$y(x) = A \sinh x \quad (0 \leq x < \pi/2), \quad y(x) = C \sinh(\pi - x) \quad (\pi/2 < x \leq \pi).$$

b) **CCJ** *Continuity Condition and Jump.* Use continuity to prove $A = C$. Integrate the equation across $[\pi/2 - \eta, \pi/2 + \eta]$, let $\eta \rightarrow 0^+$, and prove

$$y'(\pi/2^+) - y'(\pi/2^-) = 3.$$

c) **BVDC** *Boundary Values Determine the Constants.* Deduce

$$A = C = -\frac{3}{2 \cosh(\pi/2)},$$

and write the resulting piecewise solution, including its common value at $x = \pi/2$.

Indication.— The delta forcing acts through the highest derivative, so y is continuous while y' jumps; letting y itself jump would produce a derivative more singular than δ .

Heaviside step. The *Heaviside step function* is the function $H: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$H(x) = \begin{cases} 0, & x < 0, \\ 1, & x > 0, \end{cases}$$

for $x \neq 0$, with its value at 0 left unspecified. It represents an input switched on at the origin.

62)

a) **HSD** *Heaviside Step Definition.* Use DSP formally to justify $H(x) = \int_{-\infty}^x \delta(t) dt$ for $x \neq 0$.

b) **DDH** *Delta Derivative of Heaviside.* Let $a, b \in \mathbb{R}$ satisfy $a < 0 < b$, and let $g: [a, b] \rightarrow \mathbb{R}$ be continuously differentiable with $g(a) = g(b) = 0$. Prove by integration by parts on $[a, 0)$ and $(0, b]$ that

$$-\int_a^b H(x)g'(x) dx = g(0).$$

Deduce the generalized identity $H' = \delta$ by comparison with DSP.

End of Part VI

Part VII. — Series solutions near ordinary and singular points

This part makes the formal power-series method systematic. The convergence and existence theorems stated in the original notes remain admitted results; the coefficient calculations themselves are developed in the questions.

A. — Classification of expansion points

Analytic functions. Let $I \subseteq \mathbb{R}$ be an open interval and let $x_0 \in I$. A function $h: I \rightarrow \mathbb{R}$ is *analytic at x_0* if there are $\rho > 0$ and coefficients $(a_n)_{n \in \mathbb{N}_0}$ such that $(x_0 - \rho, x_0 + \rho) \subseteq I$ and $h(x) = \sum_{n=0}^{\infty} a_n(x - x_0)^n$ throughout that interval, with the series convergent there.

Second-order equations at a point. Let $I \subseteq \mathbb{R}$ be an open interval, let $x_0 \in I$, let $p, q, r: I \rightarrow \mathbb{R}$, and suppose that p is non-zero on some punctured neighbourhood of x_0 . Consider

$$p(x)y'' + q(x)y' + r(x)y = 0$$

for an unknown twice differentiable function y .

Ordinary points. The point x_0 is an *ordinary point* of the equation above if q/p and r/p , initially defined on the punctured neighbourhood, extend to functions analytic at x_0 .

Singular points. The point x_0 is a *singular point* of the equation if it is not an ordinary point. A zero of p is therefore only a candidate singular point: the analyticity of the two quotients is what decides.

Regular singular points. A singular point x_0 is *regular singular* if $(x - x_0)q(x)/p(x)$ and $(x - x_0)^2r(x)/p(x)$ extend analytically to x_0 . This is equivalent to the $P(x)(x - x_0)^2y'' + Q(x)(x - x_0)y' + R(x)y = 0$ form used in the source.

Irregular singular points. A singular point that is not regular singular is *irregular singular*.

63)

- a) **OPVC** *Ordinary Point with a Vanishing Coefficient.* Give an example of an equation $p(x)y'' + q(x)y' + r(x)y = 0$ with $p(0) = 0$ for which 0 is nevertheless an ordinary point, and verify the required analytic extensions.
- b) **RSPT** *Regular Singularity through Pole Tests.* Prove that a singular point x_0 is regular singular precisely when there are functions A, B analytic at x_0 with

$$\frac{q(x)}{p(x)} = \frac{A(x)}{x - x_0}, \quad \frac{r(x)}{p(x)} = \frac{B(x)}{(x - x_0)^2}$$

throughout a punctured neighbourhood of x_0 .

- c) **CPE** *Classification of Principal Examples.* Determine all ordinary, regular singular, and irregular singular points of each equation:

$$\begin{aligned} (1 - x^2)y'' - 2xy' + 2y &= 0, \\ (\sin x)y'' + (\cos x)y' + 2y &= 0, \\ (1 + \sqrt{x})y'' - 2xy' + 2y &= 0 \quad (x \geq 0). \end{aligned}$$

In particular, prove that 0 is ordinary and ± 1 are regular singular in the first equation; every $n\pi$, $n \in \mathbb{Z}$, is regular singular in the second; and 0 is irregular singular in the third because the required coefficients are not analytic there.

Admitted result. **OPSB** *Ordinary-Point Series Basis.* Let $I \subseteq \mathbb{R}$ be an open interval, let $x_0 \in I$, and let $P, Q: I \rightarrow \mathbb{R}$ be analytic at x_0 . There is an open interval $J \subseteq I$

containing x_0 on which $y'' + P(x)y' + Q(x)y = 0$ has two linearly independent solutions represented by convergent power series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$. Equivalently, for every $u_0, v_0 \in \mathbb{R}$ there is exactly one such local analytic solution satisfying $y(x_0) = u_0$ and $y'(x_0) = v_0$.

Frobenius series. At a regular singular point x_0 , the *Frobenius method* substitutes

$$y = \sum_{n=0}^{\infty} a_n(x - x_0)^{n+\sigma}, \quad a_0 \neq 0.$$

Indicial equation. In the Frobenius substitution, the lowest-power coefficient gives the *indicial equation* for the *index* σ .

Admitted result. FRSE Frobenius Regular-Singularity Existence. Let $I \subseteq \mathbb{R}$ be an open interval, let $x_0 \in I$, and let $p, q, r: I \rightarrow \mathbb{R}$ be such that x_0 is a regular singular point of $p(x)y'' + q(x)y' + r(x)y = 0$. There exist $\rho > 0$, an index $\sigma \in \mathbb{C}$, and coefficients $(a_n)_{n \in \mathbb{N}_0}$ in $\mathbb{C}$, with $a_0 \neq 0$ and $(x_0, x_0 + \rho) \subseteq I$, for which

$$y(x) = (x - x_0)^\sigma \sum_{n=0}^{\infty} a_n(x - x_0)^n,$$

defines a non-zero complex-valued solution for $x_0 < x < x_0 + \rho$. The power series converges for $|x - x_0| < \rho$, and σ is a root of the indicial equation.

B. — Ordinary-point recurrences

64) Seek a power-series solution about 0 of

$$(1 - x^2)y'' - 2xy' + 2y = 0$$

in the form $y = \sum_{n=0}^{\infty} a_n x^n$.

a) **OPSS Ordinary-Point Series Substitution.** Write series for y' and y'' , align their powers of x , and prove that the coefficient of x^n vanishes precisely when

$$(n+2)(n+1)a_{n+2} + (2 - n(n+1))a_n = 0 \quad (n \in \mathbb{N}_0).$$

b) **LCR Legendre Coefficient Recurrence.** Deduce $a_{n+2} = (n-1)a_n/(n+1)$. Treat $n = 0$ and $n = 1$ before making any division, and show that a_0 and a_1 may be chosen arbitrarily while the recurrence determines every other coefficient.

c) **EOA Even and Odd Alternatives.** Prove that the odd branch terminates at a_1x , while

$$a_{2k} = -\frac{a_0}{2k-1} \quad (k \in \mathbb{N}).$$

Hence obtain

$$y = a_0 \left(1 - x^2 - \frac{x^4}{3} - \frac{x^6}{5} - \cdots \right) + a_1 x.$$

d) **LSF Logarithmic Summation Form.** For $|x| < 1$, use

$$\log \left(\frac{1+x}{1-x} \right) = 2 \sum_{k=0}^{\infty} \frac{x^{2k+1}}{2k+1}$$

to sum the even branch and prove

$$y = a_0 \left[1 - \frac{x}{2} \log \left(\frac{1+x}{1-x} \right) \right] + a_1 x.$$

Show that the even branch has logarithmic singularities at $x = \pm 1$, the regular singular points of the equation.

C. — Frobenius indices and resonance

65) Consider

$$4xy'' + 2(1 - x^2)y' - xy = 0$$

on $(0, \infty)$ near 0.

- a) **RSC** *Regular Singularity Check*. Multiply by x and show that the normalized coefficients required in the regular-singular definition are $(1 - x^2)/2$ and $-x^2/4$.
- b) **GFR** *General Frobenius Recurrence*. Substitute $y = \sum_{n=0}^{\infty} a_n x^{n+\sigma}$, using $a_{-2} = a_{-1} = 0$, and derive

$$2(n + \sigma)(2n + 2\sigma - 1)a_n = (2n + 2\sigma - 3)a_{n-2} \quad (n \in \mathbb{N}_0).$$

- c) **IRI** *Indicial Roots Identification*. Set $n = 0$ before dividing and prove that $2\sigma(2\sigma - 1)a_0 = 0$, so $\sigma = 0$ or $\sigma = 1/2$.
- d) **ASB** *Analytic Series Branch*. For $\sigma = 0$, prove $a_1 = 0$ and

$$a_n = \frac{2n - 3}{2n(2n - 1)} a_{n-2} \quad (n \geq 2).$$

Calculate the coefficients of x^2 and x^4 and write the analytic branch

$$y = a_0 \left(1 + \frac{x^2}{12} + \frac{5x^4}{672} + \cdots \right).$$

- e) **HIB** *Half-Integer Branch*. For $\sigma = 1/2$, prove $a_1 = 0$ and

$$a_n = \frac{n - 1}{n(2n + 1)} a_{n-2} \quad (n \geq 2).$$

Calculate the first two non-constant coefficients and obtain the branch $x^{1/2}(1 + x^2/10 + x^4/120 + \cdots)$ up to an arbitrary non-zero factor.

Admitted result. **FSRD** *Frobenius Solutions by Root Difference*. Let $I \subseteq \mathbb{R}$ be an open interval, let $x_0 \in I$, and let $p, q, r: I \rightarrow \mathbb{R}$ be such that x_0 is a regular singular point of $p(x)y'' + q(x)y' + r(x)y = 0$. Let $\sigma_1, \sigma_2 \in \mathbb{C}$ be the roots of its indicial equation, ordered so that $\operatorname{Re} \sigma_1 \leq \operatorname{Re} \sigma_2$. If $\sigma_2 - \sigma_1$ is not an integer, two linearly independent Frobenius series solutions arise from the two roots on a sufficiently small interval to the right of x_0 . If the difference is an integer, including zero, a Frobenius series solution y_1 exists for the larger root and a second independent solution generally has the form

$$y_2 = C \log(x - x_0)y_1 + (x - x_0)^{\sigma_1} \sum_{n=0}^{\infty} b_n (x - x_0)^n.$$

Here $C \in \mathbb{C}$, the coefficients $b_n \in \mathbb{C}$, and the displayed series converges on that interval. The constant C can vanish, in which case two log-free Frobenius solutions may exist even though the root difference is an integer.

66) Consider $x^2y'' - xy = 0$ on $(0, \infty)$ near 0.

- a) **RFRE** *Resonant Frobenius Recurrence Equation*. Substitute $y = \sum_{n=0}^{\infty} a_n x^{n+\sigma}$, with $a_{-1} = 0$, and derive

$$(n + \sigma)(n + \sigma - 1)a_n = a_{n-1} \quad (n \in \mathbb{N}_0).$$

Obtain the indicial roots 0 and 1.

- b) **URB** *Upper Root Branch*. For $\sigma = 1$, prove $a_n = a_0/((n + 1)(n!)^2)$ and hence

$$y_1 = a_0 x \left(1 + \frac{x}{2} + \frac{x^2}{12} + \frac{x^3}{144} + \cdots \right).$$

- c) **LRF** *Lower Root Failure.* For $\sigma = 0$, inspect the recurrence first at $n = 0$ and then at $n = 1$. Prove that it would require $0 \cdot a_1 = a_0$, contradicting $a_0 \neq 0$. Show that setting $a_0 = 0$ in this branch reproduces the upper-root series.
- d) **LSS** *Logarithmic Second Solution.* Apply FSRD and the failure in part (c) to prove that the logarithmic coefficient in a second independent solution is non-zero. After rescaling that solution, write the permitted form

$$y_2 = y_1 \log x + \sum_{n=0}^{\infty} b_n x^n,$$

and show that substituting this form into the equation determines the coefficients b_n recursively.

End of Part VII

Part VIII. — Gradients, Taylor forms, and stationary points

This part returns to functions of several variables. It extracts directional information from the total differential and uses the second-order term to classify stationary points.

A. — Gradients and directional derivatives

Euclidean dot product. For a positive integer $n \in \mathbb{N}$ and vectors $u, v \in \mathbb{R}^n$, write $u = (u_1, \dots, u_n)$ and $v = (v_1, \dots, v_n)$, and define

$$u \cdot v = \sum_{j=1}^n u_j v_j.$$

Euclidean norm. The *norm* of a vector $u \in \mathbb{R}^n$ is $|u| = \sqrt{u \cdot u}$.

Unit vectors. A vector $s \in \mathbb{R}^n$ is a *unit vector* if $|s| = 1$.

Gradient. Let $U \subseteq \mathbb{R}^n$ be open and let $f: U \rightarrow \mathbb{R}$ have first partial derivatives. Its *gradient* is the function $\nabla f: U \rightarrow \mathbb{R}^n$ defined by

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}(x), \dots, \frac{\partial f}{\partial x_n}(x) \right).$$

It is also written $\text{grad } f$.

Directional derivatives. For $x_0 \in U$ and a unit vector $s \in \mathbb{R}^n$, the *directional derivative* of f is

$$D_s f(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + hs) - f(x_0)}{h},$$

provided the limit exists.

Admitted result. CSI Cauchy–Schwarz Inequality. For every $n \in \mathbb{N}$ and all $u, v \in \mathbb{R}^n$,

$$|u \cdot v| \leq |u| |v|,$$

with equality if and only if u and v are linearly dependent.

67) Let $f: U \rightarrow \mathbb{R}$ be differentiable on U , and fix $x_0 \in U$.

- a) **TDGGV** *Total Differential Gives the Gradient Vector.* For an infinitesimal displacement $ds = (dx_1, \dots, dx_n)$, rewrite the total differential as

$$df = ds \cdot \nabla f(x_0).$$

- b) **DDGL** *Directional Derivative as a Gradient Limit.* Substitute the displacement hs in the linear approximation and prove

$$D_s f(x_0) = s \cdot \nabla f(x_0).$$

- c) **MRG** *Maximum Rate from the Gradient.* Use the Cauchy–Schwarz inequality to prove $|D_s f(x_0)| \leq |\nabla f(x_0)|$ for every unit vector s . If the gradient is non-zero, determine the directions of maximum increase and maximum decrease.
- d) **GOLC** *Gradient Orthogonality to Level Curves.* Let $I \subseteq \mathbb{R}$ be an open interval, let $\gamma: I \rightarrow U$ be differentiable, and suppose $f(\gamma(t)) = c$ for every $t \in I$, where $c \in \mathbb{R}$. Differentiate along the path and prove $\gamma'(t) \cdot \nabla f(\gamma(t)) = 0$. Deduce that the gradient is orthogonal to $\gamma'(t)$ at every t for which $\gamma'(t) \neq 0$.

Stationary points. A point $x_0 \in U$ is a *stationary point*, or *critical point*, of f if $\nabla f(x_0) = 0$.

Local extrema. A stationary point x_0 is a *local minimum* if there is a radius $\rho > 0$ such that $f(x) \geq f(x_0)$ whenever $x \in U$ and $|x - x_0| < \rho$, and a *strict local minimum* if the inequality is strict for $x \neq x_0$. It is a *local maximum* if the same condition holds with $f(x) \leq f(x_0)$, with the analogous *strict* variant.

Saddle points. A *saddle point* is a stationary point that is neither a local minimum nor a local maximum.

68)

- a) **AVSD** *All Vanishing Stationary Directions.* Let $f: U \rightarrow \mathbb{R}$ be differentiable at a stationary point $x_0 \in U$. Prove that every directional derivative of f at x_0 vanishes.
- b) **SDD** *Saddle Direction Demonstration.* For $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $f(x, y) = x^2 - y^2$, prove that $(0, 0)$ is stationary and exhibit directions along which f increases and decreases. Prove that the zero contour consists of the two crossing lines $y = x$ and $y = -x$.

B. — The Hessian and multivariable Taylor expansion

Hessian matrix. Let $U \subseteq \mathbb{R}^n$ be open and let $f: U \rightarrow \mathbb{R}$ have continuous second partial derivatives. Its *Hessian matrix* is the function $H_f: U \rightarrow \mathbb{R}^{n \times n}$ defined at $x \in U$ by

$$H_f(x) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(x) \right)_{1 \leq i, j \leq n}.$$

By SCECP it is a real symmetric matrix.

Eigenvectors and eigenvalues. Let $n \in \mathbb{N}$ and let M be a real $n \times n$ matrix. A vector $v \in \mathbb{C}^n \setminus \{0\}$ is an *eigenvector* of M with *eigenvalue* $\lambda \in \mathbb{C}$ if $Mv = \lambda v$; the eigenvalues are the roots of $\det(M - \lambda I_n) = 0$, where I_n is the $n \times n$ identity matrix.

Non-degenerate stationary points. A stationary point x_0 is *non-degenerate* if $H_f(x_0)$ is invertible, that is, if zero is not an eigenvalue of $H_f(x_0)$.

69) Let $x_0 \in U$ and $h \in \mathbb{R}^n$ be sufficiently small that $x_0 + th \in U$ for every $t \in [0, 1]$. Define $g: [0, 1] \rightarrow \mathbb{R}$ by $g(t) = f(x_0 + th)$.

- a) **LPR** *Line Parameter Restriction.* Apply the multivariable chain rule to prove $g'(t) = h \cdot \nabla f(x_0 + th)$.
- b) **LRSD** *Line-Restricted Second Derivative.* Differentiate once more and prove

$$g''(t) = h^\top H_f(x_0 + th)h = \sum_{i=1}^n \sum_{j=1}^n h_i h_j \frac{\partial^2 f}{\partial x_i \partial x_j}(x_0 + th).$$

- c) **SOTQ** *Second-Order Term as a Quadratic form.* In dimension two, with $h = (\delta x, \delta y)$, expand $h^\top H_f(x_0)h$ as

$$f_{xx}(\delta x)^2 + 2f_{xy}\delta x\delta y + f_{yy}(\delta y)^2,$$

where all derivatives are evaluated at x_0 .

- d) **MTE** *Multivariable Taylor Expansion.* Apply the Fundamental Theorem of Calculus twice to prove

$$g(1) = g(0) + g'(0) + \int_0^1 (1-t)g''(t) dt.$$

Use the continuity of the Hessian at x_0 to deduce

$$f(x_0 + h) = f(x_0) + h \cdot \nabla f(x_0) + \frac{1}{2}h^\top H_f(x_0)h + o(|h|^2)$$

as $h \rightarrow 0$.

C. — Classification and contours

Quadratic forms. For a real symmetric $n \times n$ matrix H , its *quadratic form* is the function $Q_H: \mathbb{R}^n \rightarrow \mathbb{R}$ defined by $Q_H(h) = h^T H h$.

Positive definite quadratic forms. The form Q_H is *positive definite* if $Q_H(h) > 0$ for every $h \neq 0$.

Negative definite quadratic forms. The form Q_H is *negative definite* if $Q_H(h) < 0$ for every $h \neq 0$.

Indefinite quadratic forms. The form Q_H is *indefinite* if it takes both positive and negative values.

Admitted result. SSTD Symmetric Spectral Theorem and Diagonalization. Let $n \in \mathbb{N}$. For every real symmetric $n \times n$ matrix H , there is an orthonormal basis of $\mathbb{R}^n$ (a basis of mutually orthogonal unit vectors) consisting of eigenvectors of H . In the corresponding coordinates, H is diagonal with real eigenvalues $\lambda_1, \dots, \lambda_n$, and

$$h^T H h = \sum_{j=1}^n \lambda_j X_j^2,$$

where $X_1, \dots, X_n$ are the coordinates of h in that basis.

70) Let x_0 be a stationary point of a function $f: U \rightarrow \mathbb{R}$ having the Taylor expansion from Question 69.

- a) **QFDC Quadratic Form Definiteness Cases.** Use SSTD to prove that a positive-definite Hessian satisfies $h^T H_f(x_0) h \geq m|h|^2$ for some $m > 0$ and every $h \in \mathbb{R}^n$. Use the remainder definition to deduce a strict local minimum. Give the corresponding argument for a negative-definite Hessian and a strict local maximum.
- b) **SEM Spectral Eigenvalue Method.** Apply SSTD to prove that the Hessian is positive definite exactly when all its eigenvalues are positive, negative definite exactly when they are all negative, and indefinite when it has eigenvalues of both signs.
- c) **ISPC Indefinite Sign Produces a Crossing.** If the Hessian is indefinite, use two eigenvector directions and the Taylor expansion to prove that x_0 is a saddle point.
- d) **ZEC Zero Eigenvalue Caveat.** Show that the one-variable functions $x \mapsto x^4$, $x \mapsto -x^4$, and $x \mapsto x^3$ all have vanishing Hessian at 0 but have respectively a strict local minimum, a strict local maximum, and a saddle there, so that a zero eigenvalue leaves the classification undetermined.

Leading principal minors. Let $n \in \mathbb{N}$. For a real symmetric $n \times n$ matrix H , the k th *leading principal minor* Δ_k is the determinant of its upper-left $k \times k$ submatrix, for $1 \leq k \leq n$.

Admitted result. SCFD Sylvester Conditions For Definiteness. Let $n \in \mathbb{N}$ and let H be a real symmetric $n \times n$ matrix with leading principal minors $\Delta_1, \dots, \Delta_n$. The matrix is positive definite if and only if $\Delta_k > 0$ for every $1 \leq k \leq n$. It is negative definite if and only if $(-1)^k \Delta_k > 0$ for every $1 \leq k \leq n$. Failure of both tests need not imply indefiniteness when zero eigenvalues are present.

71)

- a) **LPMD Leading Principal Minor Determinants.** For $H = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$, compute $\Delta_1 = a$ and $\Delta_2 = ac - b^2$.
- b) **SCPA Sylvester Criterion Positive Application.** Use SCFD to prove that H is positive definite precisely when $a > 0$ and $ac - b^2 > 0$.

- c) **SNA** *Sylvester Negative Application*. Prove the corresponding negative-definite conditions $a < 0$ and $ac - b^2 > 0$.
- d) **ITD** *Indefiniteness in Two Dimensions*. Prove that $ac - b^2 < 0$ makes H indefinite. When $ac - b^2 = 0$, give examples of two functions with the same quadratic second-order term at a stationary point but different local behaviour there.

72) Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = 4x^3 - 12xy + y^2 + 10y + 6.$$

- a) **SPS** *Solving Polynomial Stationarity*. Compute the gradient and prove that the stationary points are exactly $(1, 1)$ and $(5, 25)$.
- b) **HCPE** *Hessian Computation at Principal Examples*. Prove

$$H_f(x, y) = \begin{pmatrix} 24x & -12 \\ -12 & 2 \end{pmatrix}.$$

At $(1, 1)$ compute a negative determinant and deduce that the point is a saddle.
 At $(5, 25)$ compute leading principal minors 120 and 96 and deduce that the point is a strict local minimum.

- c) **QCC** *Quadratic Contour Classification*. In coordinates aligned with the Hessian eigenvectors, show that the quadratic-level curves near a non-degenerate minimum or maximum are ellipses, while those near a saddle are hyperbolas. Prove that the saddle's quadratic zero contour consists of two lines crossing at the stationary point.

End of Part VIII

Part IX. — Differential systems and phase portraits

This part packages interacting first-order equations in vector form. It connects eigenvalues with phase portraits and then applies the linear theory locally to nonlinear population dynamics.

A. — Linear systems and eigenmodes

Matrix-vector products. For a positive integer $n \in \mathbb{N}$, let $M = (m_{ij})_{1 \leq i, j \leq n}$ be a real $n \times n$ matrix and let $Y = (y_1, \dots, y_n)^T \in \mathbb{R}^n$. Define $(MY)_i = \sum_{j=1}^n m_{ij}y_j$.

Trace. The *trace* of the matrix above is $\text{tr } M = \sum_{i=1}^n m_{ii}$.

Two-dimensional linear systems. For the two-dimensional system, let $I \subseteq \mathbb{R}$ be an open interval, let $a, b, c, d \in \mathbb{R}$, let $f_1, f_2: I \rightarrow \mathbb{R}$ be continuous, and let the unknown functions $y_1, y_2: I \rightarrow \mathbb{R}$ be differentiable. Write

$$\dot{Y} = MY + F(t), \quad Y(t) = \begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix}, \quad M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad F(t) = \begin{pmatrix} f_1(t) \\ f_2(t) \end{pmatrix}.$$

The dot denotes differentiation with respect to $t \in I$.

73)

- a) **ESOE** *Elimination gives a Second-Order Equation.* Assume f_1 is differentiable. Differentiate $\dot{y}_1 = ay_1 + by_2 + f_1$, substitute both system equations, and prove

$$\ddot{y}_1 - (a + d)\dot{y}_1 + (ad - bc)y_1 = bf_2 - df_1 + \dot{f}_1.$$

Do not divide by b , so the identity remains valid when $b = 0$.

- b) **SOFS** *Second Order as a First-order System.* Let $\alpha, \beta \in \mathbb{R}$, let $f: I \rightarrow \mathbb{R}$, and let $y: I \rightarrow \mathbb{R}$ be twice differentiable. By setting $y_1 = y$ and $y_2 = \dot{y}$, transform $\ddot{y} + \alpha\dot{y} + \beta y = f$ into

$$\dot{Y} = \begin{pmatrix} 0 & 1 \\ -\beta & -\alpha \end{pmatrix} Y + \begin{pmatrix} 0 \\ f \end{pmatrix}.$$

- c) **SMN** *System Matrix Notation.* Expand the vector equation component by component and verify that it is exactly equivalent to the original pair of scalar equations.

Admitted result. **LSIVPEU** *Linear System Initial-Value Problem Existence and Uniqueness.* Let $n \in \mathbb{N}$, let $I \subseteq \mathbb{R}$ be an open interval, let $A: I \rightarrow \mathbb{R}^{n \times n}$ and $F: I \rightarrow \mathbb{R}^n$ be continuous, let $t_0 \in I$, and let $Y_0 \in \mathbb{R}^n$. There exists a unique continuously differentiable function $Y: I \rightarrow \mathbb{R}^n$ satisfying $\dot{Y}(t) = A(t)Y(t) + F(t)$ for every $t \in I$ and $Y(t_0) = Y_0$.

74) Let $n \in \mathbb{N}$ and let M be a constant real $n \times n$ matrix. Seek complex-valued solutions on $\mathbb{R}$.

- a) **EAS** *Eigenvector Ansatz for a System.* Let $\lambda \in \mathbb{C}$ and $v \in \mathbb{C}^n \setminus \{0\}$. Substitute $Y(t) = ve^{\lambda t}$ in $\dot{Y} = MY$ and prove that it is a non-zero solution precisely when $Mv = \lambda v$.
- b) **DES** *Distinct Eigenvalue Solutions.* Suppose M has n distinct complex eigenvalues $\lambda_1, \dots, \lambda_n$ with eigenvectors $v_1, \dots, v_n$. Prove, by induction on the number of vectors and an application of $M - \lambda_n I_n$, that these eigenvectors are linearly independent. Show that, for $A_1, \dots, A_n \in \mathbb{C}$,

$$Y(t) = \sum_{j=1}^n A_j v_j e^{\lambda_j t}$$

realizes arbitrary initial data in $\mathbb{C}^n$. Apply LSIVPEU to the real and imaginary parts to conclude that this is the general complex solution.

- c) **RCMC** *Real Combinations from Matched Conjugates.* When M is real and a non-real eigenpair is (λ, v) , prove that $(\bar{\lambda}, \bar{v})$ is also an eigenpair and prove that the real and imaginary parts of $ve^{\lambda t}$ are real solutions.

75) Let

$$M = \begin{pmatrix} -4 & 24 \\ 1 & -2 \end{pmatrix}, \quad g = \begin{pmatrix} 4 \\ 1 \end{pmatrix},$$

and consider $\dot{Y} = MY + ge^t$ on $\mathbb{R}$.

- a) **CEC** *Coupled Eigenvalue Calculation.* Compute $\det(M - \lambda I)$ and obtain the eigenvalues 2 and -8 .
 b) **CED** *Coupled Eigenvector Directions.* Find corresponding eigenvectors $v_1 = (4, 1)^\top$ and $v_2 = (-6, 1)^\top$.
 c) **EFPI** *Exponential Forcing Particular Integral.* Seek $Y_p = ue^t$, solve $(I_2 - M)u = g$, and prove $u = (-4, -1)^\top$.
 d) **CGS** *Complete General Solution.* Deduce

$$Y(t) = Av_1e^{2t} + Bv_2e^{-8t} - \begin{pmatrix} 4 \\ 1 \end{pmatrix} e^t, \quad A, B \in \mathbb{R}.$$

- e) **SSP** *Saddle State Portrait.* For the homogeneous system, prove that trajectories on the v_2 line approach the origin and those on the v_1 line recede, and deduce the saddle geometry of the remaining trajectories from the general solution.

Admitted result. **MDFE** *Matrix Determinant Flow Equation.* Let $n \in \mathbb{N}$, let $I \subseteq \mathbb{R}$ be an open interval, let $A: I \rightarrow \mathbb{R}^{n \times n}$ be continuous, and let $\Phi: I \rightarrow \mathbb{R}^{n \times n}$ be differentiable and satisfy $\Phi' = A\Phi$. Then $W = \det \Phi$ satisfies $W' = (\operatorname{tr} A)W$. Consequently, for $t, t_0 \in I$,

$$W(t) = W(t_0) \exp \left(\int_{t_0}^t \operatorname{tr} A(s) \, ds \right).$$

In particular, W is either identically zero or nowhere zero.

76)

- a) **TAFC** *Trace Abel Formula Connection.* Let $I \subseteq \mathbb{R}$ be an open interval and let $p, q: I \rightarrow \mathbb{R}$ be continuous. Apply MDFE to the first-order form of $y'' + p(x)y' + q(x)y = 0$ and recover the Abel formula from Question 49.
 b) **IBFS** *Independence of a Basis of Fundamental Solutions.* Let $n \in \mathbb{N}$, let $I \subseteq \mathbb{R}$ be an open interval, let $A: I \rightarrow \mathbb{R}^{n \times n}$ be continuous, and let the columns of Φ be n solutions of $Y' = A(t)Y$. Prove that if they form a basis at one time $t_0 \in I$, they form a basis at every $t \in I$.

B. — Classification of linear phase portraits

Eigendirections. The line spanned by an eigenvector is called an *eigendirection*.

Saddles. For the linear system $\dot{Y} = MY$, the origin is a *saddle* when there are two real eigendirections, one along which nearby trajectories approach the origin and one along which they recede.

Nodes. The origin is a *node* when every nearby trajectory approaches or leaves the origin without rotating.

Spirals. The origin is a *spiral* when every nearby non-zero trajectory winds around the origin with alternating angular motion.

Centres. The origin is a *centre* when every nearby non-zero trajectory is a closed curve surrounding the origin.

- 77)** Let M be a real 2×2 matrix with two distinct eigenvalues, and consider $\dot{Y} = MY$.
- a) **ROSS** *Real Opposite Signs give a Saddle.* Suppose the eigenvalues are real and satisfy $\lambda_1 > 0 > \lambda_2$. Use the eigenmode solution to prove that the origin is a saddle, attracting along the λ_2 eigendirection and repelling along the λ_1 eigendirection.
 - b) **RSSN** *Real Same Signs give a Node.* Suppose the eigenvalues are real and have the same sign. Show that the origin is a stable node when both are negative and an unstable node when both are positive. Determine which eigendirection the trajectories become tangent to far from the origin.
 - c) **CCSP** *Complex Conjugates give a Spiral Portrait.* Suppose the eigenvalues are $\alpha \pm i\beta$ with $\beta \neq 0$. Use Question 74 to show that real trajectories combine a rotation of angular frequency $|\beta|$ with the factor $e^{\alpha t}$. Determine whether the spiral is inward or outward from the sign of α .
 - d) **ZRCC** *Zero Real Component gives a Centre.* If $\alpha = 0$ and the matrix is diagonalizable over $\mathbb{C}$, show that non-zero trajectories are closed curves surrounding the origin; after a real linear coordinate change they are ellipses. This case is called a centre.
 - e) **SOC** *Spiral Orientation Check.* For a spiral with eigenvalues $\alpha \pm i\beta$, $\beta \neq 0$, compute MY at the point $(1, 0)$ and deduce from the sign of its second component whether the spiral turns clockwise or counter-clockwise.

C. — Nonlinear autonomous systems

Autonomous systems. Let $D \subseteq \mathbb{R}^2$ be open and let $f, g: D \rightarrow \mathbb{R}$ have continuous first partial derivatives. An *autonomous system* on D is a pair of differential equations

$$\dot{x} = f(x, y), \quad \dot{y} = g(x, y)$$

for unknown differentiable functions x, y of t .

Equilibria of autonomous systems. An *equilibrium*, or *fixed point*, of the autonomous system is a point $(x_0, y_0) \in D$ at which $f(x_0, y_0) = g(x_0, y_0) = 0$.

Jacobian matrix. For the autonomous system above, the *Jacobian matrix* is the function $J: D \rightarrow \mathbb{R}^{2 \times 2}$ defined at $(x_0, y_0) \in D$ by

$$J(x_0, y_0) = \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix}_{(x_0, y_0)}.$$

Linearised systems. At an equilibrium (x_0, y_0) of $\dot{x} = f(x, y)$, $\dot{y} = g(x, y)$, the *linearised system* is

$$\begin{pmatrix} \dot{\xi} \\ \dot{\eta} \end{pmatrix} = J(x_0, y_0) \begin{pmatrix} \xi \\ \eta \end{pmatrix}.$$

Its eigenvalues are the candidates controlling local stability, as HLP below makes precise.

- 78)** Let (x_0, y_0) be an equilibrium and put $x = x_0 + \xi$, $y = y_0 + \eta$.

- a) **PTE** *Perturbation Taylor Expansion.* Apply the first-order multivariable expansion to prove

$$\begin{pmatrix} \dot{\xi} \\ \dot{\eta} \end{pmatrix} = J(x_0, y_0) \begin{pmatrix} \xi \\ \eta \end{pmatrix} + R(\xi, \eta),$$

where R is defined near $(0, 0)$ with values in $\mathbb{R}^2$ and

$$\frac{|R(\xi, \eta)|}{\sqrt{\xi^2 + \eta^2}} \longrightarrow 0 \quad \text{as } (\xi, \eta) \rightarrow (0, 0).$$

Admitted result. HLP Hyperbolic Linearization Principle. Let $n \in \mathbb{N}$, let $D \subseteq \mathbb{R}^n$ be open, let $F: D \rightarrow \mathbb{R}^n$ be continuously differentiable, and let $x_0 \in D$ satisfy $F(x_0) = 0$. If no eigenvalue of the Jacobian $DF(x_0)$ has zero real part, then in a sufficiently small neighbourhood the nonlinear system $\dot{x} = F(x)$ has the same qualitative equilibrium type and stability as its linearisation: if every eigenvalue has negative real part then every sufficiently close initial state converges to the equilibrium, if every eigenvalue has positive real part then every sufficiently close non-zero initial state moves away from it, and saddles retain one attracting and one repelling direction. If an eigenvalue has zero real part, this conclusion is not guaranteed and nonlinear terms must be examined.

79) Let $I \subseteq \mathbb{R}$ be an interval and let the differentiable functions $x, y: I \rightarrow [0, \infty)$ denote prey and predator populations. For parameters $\alpha, \beta, \gamma, \varepsilon, \delta > 0$, the model

$$\dot{x} = \alpha x - \beta x^2 - \gamma xy, \quad \dot{y} = \varepsilon xy - \delta y$$

combines prey growth, prey competition, predation, predator reproduction, and predator death: the term αx is prey growth, $-\beta x^2$ is prey competition, $-\gamma xy$ is predation, εxy is predator reproduction, and $-\delta y$ is predator death. Consider the particular system

$$\dot{x} = 8x - 2x^2 - 2xy, \quad \dot{y} = xy - y.$$

- a) **EEP Ecological Equilibrium Points.** Factor the two right-hand sides of the particular system and prove that its equilibria in $[0, \infty)^2$ are exactly $(0, 0)$, $(4, 0)$, and $(1, 3)$.
- b) **OJS Origin Jacobian Saddle.** Compute $J(0, 0)$, find its eigenvalues 8 and -1 , and determine the type and stability of the origin using HLP.
- c) **BJS Boundary Jacobian Saddle.** Compute $J(4, 0) = \begin{pmatrix} -8 & -8 \\ 0 & 3 \end{pmatrix}$, find eigenvectors $(1, 0)^\top$ and $(8, -11)^\top$, and determine its type.
- d) **IJS Interior Jacobian Spiral.** Compute $J(1, 3) = \begin{pmatrix} -2 & -2 \\ 3 & 0 \end{pmatrix}$ and prove that its eigenvalues are $-1 \pm i\sqrt{5}$. Deduce that $(1, 3)$ is a stable spiral.
- e) **SOD Spiral Orientation Direction.** At a perturbation $(\xi, \eta) = (\xi, 0)$ with $\xi > 0$, compute the linearised velocity $(-2\xi, 3\xi)$. Deduce that the stable spiral is counter-clockwise.

End of Part IX

Part X. — Characteristics, waves, and diffusion

This final part extends the evolution viewpoint to functions of two independent variables. It first follows information along characteristic curves, then separates left- and right-travelling waves, and finally derives a similarity profile for heat diffusion.

Partial differential equations. A *partial differential equation*, or *PDE*, is an equation involving partial derivatives of an unknown function of at least two independent variables. In this part, $x \in \mathbb{R}$ denotes position and $t \in [0, \infty)$ denotes time.

Prototype equations. The first-order wave equation is also called the advection equation; the second-order wave equation is the prototype of a hyperbolic PDE; and the diffusion equation is the prototype of a parabolic PDE. Another standard example is Laplace's equation, which models electrostatic and potential-flow fields.

A. — First-order waves and characteristics

First-order wave equation. Let $c \in \mathbb{R}$ and let $y: \mathbb{R} \times [0, \infty) \rightarrow \mathbb{R}$ have continuous first partial derivatives. The *first-order wave equation* in the sign convention of the original notes is

$$y_t - cy_x = 0.$$

Characteristics. A *characteristic* is a curve in the (x, t) -plane along which a PDE reduces to an ordinary differential equation.

80)

- a) **PCD** *Path Chain-rule Derivative.* Let $x: [0, \infty) \rightarrow \mathbb{R}$ be differentiable. Apply the chain rule to prove

$$\frac{d}{dt}y(x(t), t) = y_x(x(t), t)x'(t) + y_t(x(t), t).$$

- b) **CLCC** *Characteristic Lines Carry Constants.* Use the PDE and choose $x'(t) = -c$. Prove that $x(t) + ct$ and $y(x(t), t)$ are both constant along the resulting straight characteristics.
- c) **GSA** *General Solution of Advection.* Deduce that every continuously differentiable solution has the form

$$y(x, t) = f(x + ct)$$

for a differentiable function $f: \mathbb{R} \rightarrow \mathbb{R}$. Prove the converse by direct differentiation, and determine the direction and speed with which the profile travels.

81) In the following items, seek a continuously differentiable function $y: \mathbb{R} \times [0, \infty) \rightarrow \mathbb{R}$.

- a) **TIPA** *Translated Initial Profile under Advection.* Impose $y(x, 0) = x^2 - 3$ for every $x \in \mathbb{R}$ in the preceding equation and obtain $y(x, t) = (x + ct)^2 - 3$.
- b) **FCO** *Forced Characteristic Ordinary equation.* For

$$y_t + 5y_x = e^{-t}, \quad y(x, 0) = e^{-x^2},$$

derive the characteristics $x(t) = x_0 + 5t$ and the ordinary equation $dy/dt = e^{-t}$ along each one.

- c) **FIVP** *Forced Initial-Value Profile.* Integrate the characteristic equation, determine its constant from the initial profile, and prove

$$y(x, t) = 1 + e^{-(x-5t)^2} - e^{-t}.$$

Verify the PDE and initial condition directly.

B. — The second-order wave equation

Wave equation. Let $c \in (0, \infty)$ and let $y: \mathbb{R} \times [0, \infty) \rightarrow \mathbb{R}$ have continuous second partial derivatives. The *wave equation* is

$$y_{tt} = c^2 y_{xx}.$$

String model. For a vibrating string, $y(x, t)$ is transverse displacement and a positive mass density $\rho: \mathbb{R} \rightarrow (0, \infty)$ converts acceleration into force per unit length. In the uniform model, take $\rho(x) = \rho_0 > 0$ and the tension coefficient $K > 0$ to be constant. A straight string at rest, at any constant slope, exerts no net transverse force on an interior element; only bending produces a restoring force, so the force depends on the configuration only through the second spatial derivative y_{xx} .

Partial-differentiation operators. Write ∂_t and ∂_x for the operators of partial differentiation with respect to t and x , so that $\partial_t y = y_t$ and $\partial_x y = y_x$ for every differentiable y . For real α, β , the operator $\alpha\partial_t + \beta\partial_x$ acts by $(\alpha\partial_t + \beta\partial_x)y = \alpha y_t + \beta y_x$, and a product of operators denotes composition, the right-hand factor acting first; thus $\partial_t^2 y = y_{tt}$.

82) Let $G: \mathbb{R}^3 \rightarrow \mathbb{R}$ give the transverse restoring force per unit length of a uniform string as a function $G(y, y_x, y_{xx})$ of the local displacement and its first two spatial derivatives. Assume that adding an affine function $x \mapsto ax + b$ to y leaves the force unchanged:

$$G(y + ax + b, y_x + a, y_{xx}) = G(y, y_x, y_{xx})$$

for all $a, b, x \in \mathbb{R}$ and all real y, y_x, y_{xx} .

- a) **AIFCD** *Affine Invariance Forces Curvature Dependence.* Prove that G is independent of its first two arguments, so that the restoring force depends on the string configuration only through y_{xx} .
- b) **WESM** *Wave Equation from String Model.* Using the model $\rho_0 y_{tt} = K y_{xx}$, recover the wave equation and determine $c^2 = K/\rho_0$.

83)

- a) **OFD** *Operator Factorization by Difference.* Prove the commuting-operator identity

$$\partial_t^2 - c^2 \partial_x^2 = (\partial_t + c\partial_x)(\partial_t - c\partial_x).$$

- b) **WNC** *Wave Nullspace Components.* Verify that $f(x + ct)$ is annihilated by $\partial_t - c\partial_x$ and $g(x - ct)$ by $\partial_t + c\partial_x$, for twice differentiable $f, g: \mathbb{R} \rightarrow \mathbb{R}$.
- c) **CCT** *Characteristic Coordinate Transformation.* Put $\xi = x + ct$ and $\eta = x - ct$. Use the chain rule to prove

$$y_{tt} - c^2 y_{xx} = -4c^2 y_{\xi\eta}.$$

Verify that the coordinate change is invertible by expressing x and t in terms of ξ and η .

- d) **WDAS** *Wave Decomposition into Advancing Signals.* Integrate $y_{\xi\eta} = 0$ first with respect to one variable and then the other to prove the general form

$$y(x, t) = f(x + ct) + g(x - ct).$$

Determine which of $f(x + ct)$ and $g(x - ct)$ travels left and which travels right, and with what speed.

Data for a wave equation. On the whole real line, prescribe an initial displacement $\phi: \mathbb{R} \rightarrow \mathbb{R}$ and initial velocity $\psi: \mathbb{R} \rightarrow \mathbb{R}$. On a bounded spatial interval, the second spatial derivative additionally calls for two boundary conditions, one at each end, while the second time derivative calls for the two initial functions.

84) Assume ϕ is twice continuously differentiable and ψ is continuously differentiable.

- a) **IDR** *Initial Data Relations*. Substitute $t = 0$ in the two-wave representation and prove $f + g = \phi$ and $c(f' - g') = \psi$.
- b) **DIF** *Dual-wave Initial Formula*. Solve these relations, allowing an irrelevant opposite additive constant in f and g , and derive

$$y(x, t) = \frac{\phi(x + ct) + \phi(x - ct)}{2} + \frac{1}{2c} \int_{x-ct}^{x+ct} \psi(s) \, ds.$$

- c) **SVZ** *Symmetric Zero-Velocity waves*. When ψ is the zero function, prove that $y(x, t) = \frac{1}{2}\phi(x + ct) + \frac{1}{2}\phi(x - ct)$, so the initial profile splits into two equal waves travelling in opposite directions.
- d) **TLP** *Travelling Lorentzian Pulses*. Apply part (c) to $\phi(x) = 1/(1 + x^2)$ and prove

$$y(x, t) = \frac{1}{2} \left[\frac{1}{1 + (x + ct)^2} + \frac{1}{1 + (x - ct)^2} \right].$$

Check that $y(x, t) \rightarrow 0$ as $x \rightarrow \pm\infty$ for each fixed $t \geq 0$.

C. — Diffusion and similarity

Diffusion equation. Let $\kappa \in (0, \infty)$ be the thermal diffusivity, a positive constant with dimensions of length squared per unit time that measures how fast temperature variations spread. For a temperature $T: [0, \infty) \times (0, \infty) \rightarrow \mathbb{R}$ having continuous derivatives T_t and T_{xx} , the *diffusion equation*, or *heat equation*, is

$$T_t = \kappa T_{xx}.$$

Unlike the wave equation, it relates rate of change directly to curvature, so temperature irregularities smooth rather than propagate unchanged.

Error function. Define the *error function* $\operatorname{erf}: \mathbb{R} \rightarrow \mathbb{R}$ by

$$\operatorname{erf}(\eta) = \frac{2}{\sqrt{\pi}} \int_0^\eta e^{-u^2} \, du.$$

The Gaussian integral used in Question 60 gives $\operatorname{erf}(\eta) \rightarrow 1$ as $\eta \rightarrow \infty$.

Complementary error function. The *complementary error function* is $\operatorname{erfc}(\eta) = 1 - \operatorname{erf}(\eta)$ for every $\eta \in \mathbb{R}$. The conditions below model a bar initially at temperature zero whose end is switched to unit temperature at $t = 0$, equivalently by the Heaviside input from Question 62 when negative times are included.

85) Consider a semi-infinite bar satisfying, for $x > 0$ and $t > 0$,

$$\begin{aligned} T_t &= \kappa T_{xx}, & T(0, t) &= 1, \\ T(x, t) &\longrightarrow 0 & \text{as } t \rightarrow 0^+ \text{ for each fixed } x > 0, \\ T(x, t) &\longrightarrow 0 & \text{as } x \rightarrow \infty \text{ for each fixed } t > 0. \end{aligned}$$

- a) **DLS** *Diffusive Length Scale*. Use dimensions $[\kappa] = L^2 T^{-1}$ to show that $\sqrt{\kappa t}$ is the natural penetration length. For a twice differentiable function $\theta: (0, \infty) \rightarrow \mathbb{R}$, let the similarity variable and ansatz be defined by

$$\eta = \frac{x}{2\sqrt{\kappa t}}, \quad T(x, t) = \theta(\eta).$$

- b) **SVD** *Similarity Variable Derivatives*. Compute η_x and η_t , then prove

$$T_{xx} = \frac{1}{4\kappa t} \theta''(\eta), \quad T_t = -\frac{\eta}{2t} \theta'(\eta).$$

- c) **RSO** *Reduced Similarity Ordinary equation.* Substitute in the heat equation and obtain $\theta'' + 2\eta\theta' = 0$.
- d) **EFS** *Error Function Solution.* Treat θ' as the unknown in a first-order equation. Use the integrating factor e^{η^2} and prove $\theta(\eta) = A \operatorname{erf}(\eta) + B$ for constants $A, B \in \mathbb{R}$.
- e) **BCSS** *Boundary Conditions Select the Similarity solution.* Translate the boundary condition at $x = 0$, the far-field condition, and the one-sided initial condition for fixed $x > 0$ into conditions at $\eta = 0$ and $\eta = \infty$. Deduce

$$T(x, t) = \operatorname{erfc}\left(\frac{x}{2\sqrt{\kappa t}}\right).$$

- f) **PDS** *Penetration Distance Scaling.* Fix a temperature level $\Theta \in (0, 1)$. Show that the equation $T(x, t) = \Theta$ determines a constant similarity coordinate and hence a penetration distance $x = O(\sqrt{\kappa t})$. If a bar has length $L > 0$, derive the semi-infinite validity condition $t \ll L^2/\kappa$.

End of Part X and of the problem